



Workshop

How we learned to love the git

Who are we?



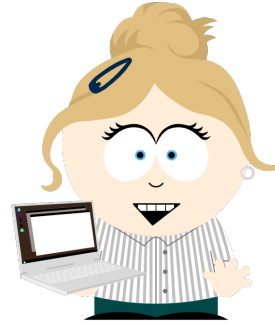
Data Scientists, SUND Data Lab, Centre for Health Data Science (HeaDS).



Post-doc, Computational Biology Laboratory, Danish cancer society Research Centre (DCRC).



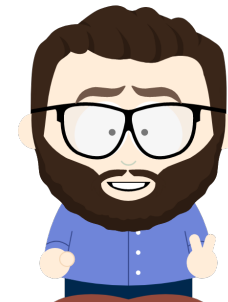
Thilde



Viki



Henrike

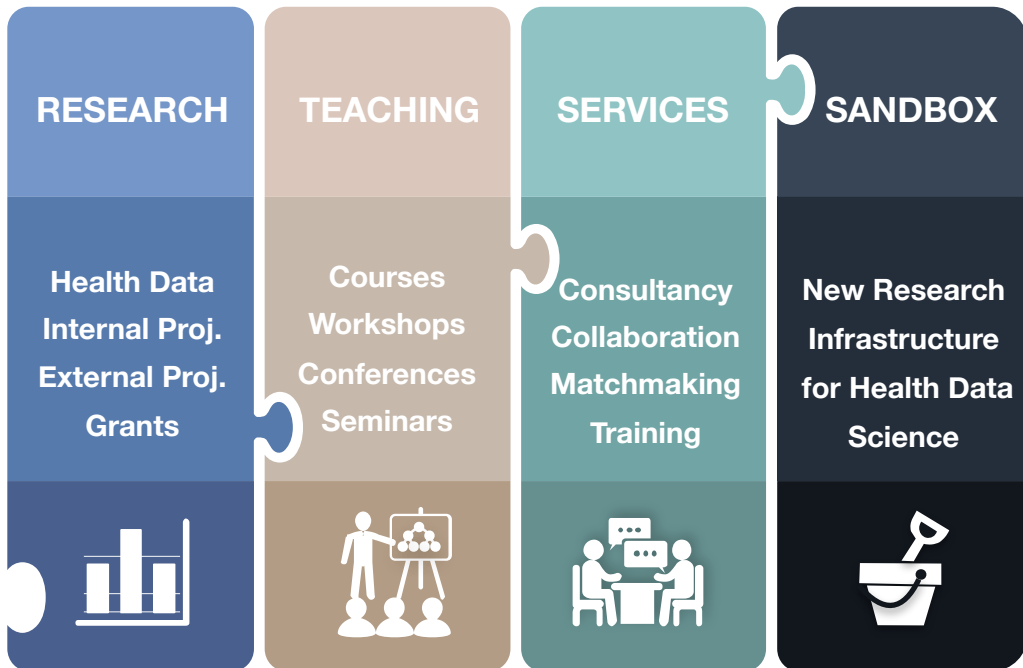


Matteo Tiberti



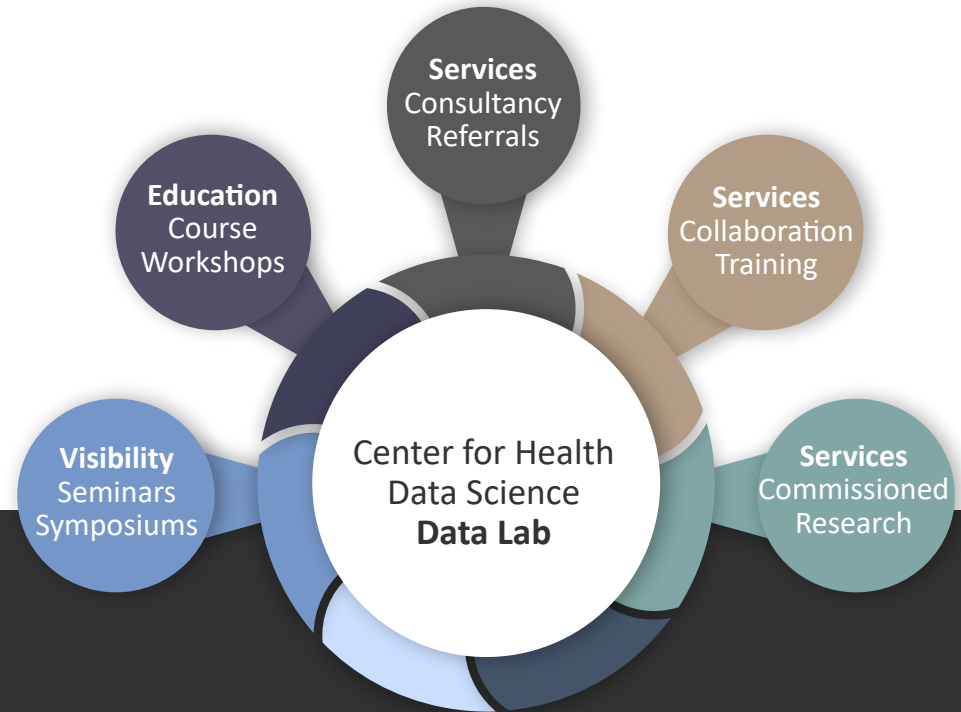
Center for Health Data Science

- **Conduct Health Data Science research.**
- **Serve as the UCPH Data Lab.**
 - Hub for health data science research.
 - Provide SUND researchers with health data science services & training.
- Develop National Health **Data Science Sandbox** for Training and Research





SUND DATA LAB



Website - <https://heads.ku.dk>

Email - datalab@sund.ku.dk

Follow us on **Twitter** at [@ucph_heads!](https://twitter.com/ucph_heads)

Slack channel and **mailing list**

courses, conferences, research activities, job postings, etc. within the field of data science.



SUND DATA LAB

COURSES

Python



R



Bash



Git



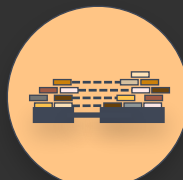
HPC



GDPR



RNA Seq

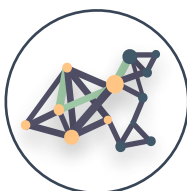


SKILLS

DEA



NW



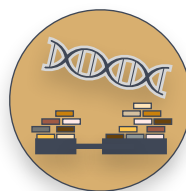
GESA



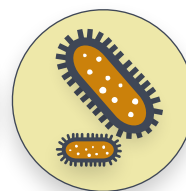
CLASS



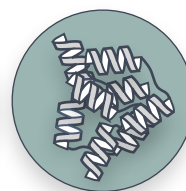
RNA



BAC.



PROTEIN



Resources

- Git Pro Book:
<https://git-scm.com/book/en/v2>



- GitHub Skills:
<https://skills.github.com>



- Git Tutorials:
<https://www.atlassian.com/git/tutorials>



- Git Cheat Sheet:
<https://education.github.com/git-cheat-sheet-education.pdf>



Program

9.00-9.45: Theoretical introduction

9.45-10.15: Create a git repository

10.15-10.30: Coffee Break

10.30-11.15: Make changes & check history

11.15-12.00: Create and merge branches

12.00-13.00: Lunch break

13.00-13.30: Go to the 'past' and fixing mistakes

13.30-14.00: Interact with a remote repository

14.00-14.30: Working together on GitHub

14.30-14.45: Coffee Break

14.45-15.30: Git & RStudio

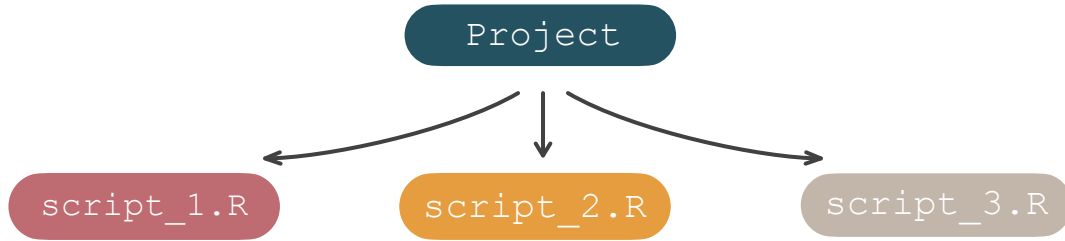
15.30-16.00: Q & A (and help with personal repo)

Justification

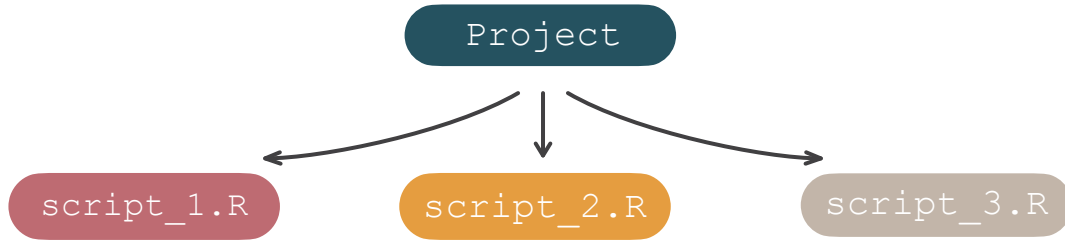
Project



Justification



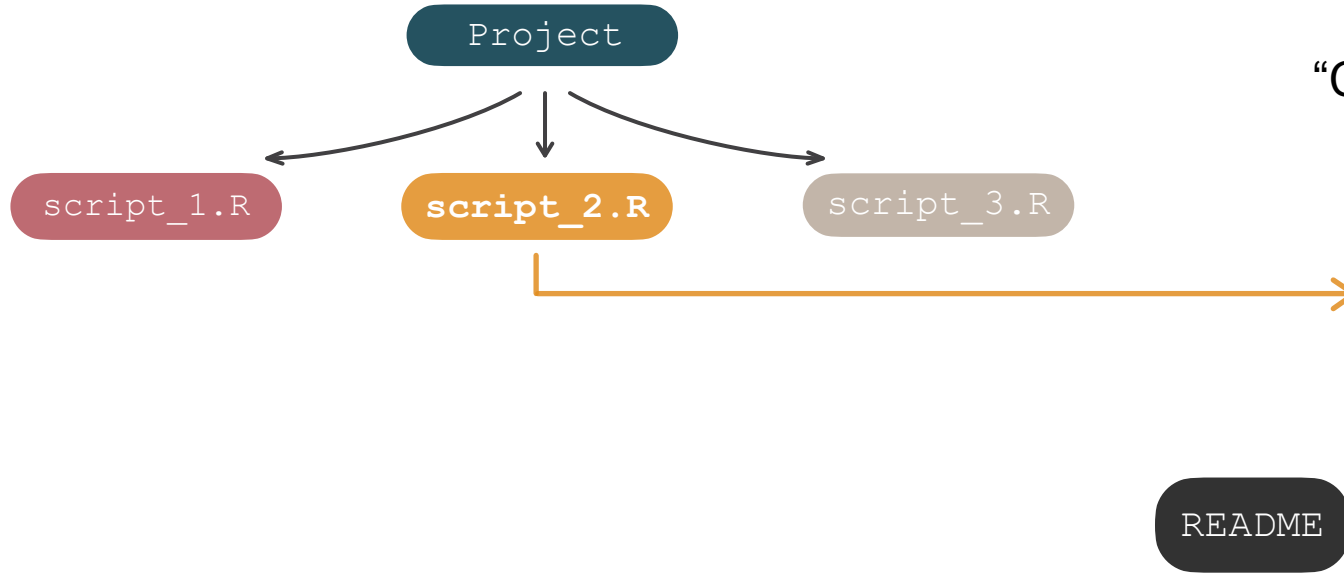
Justification



README



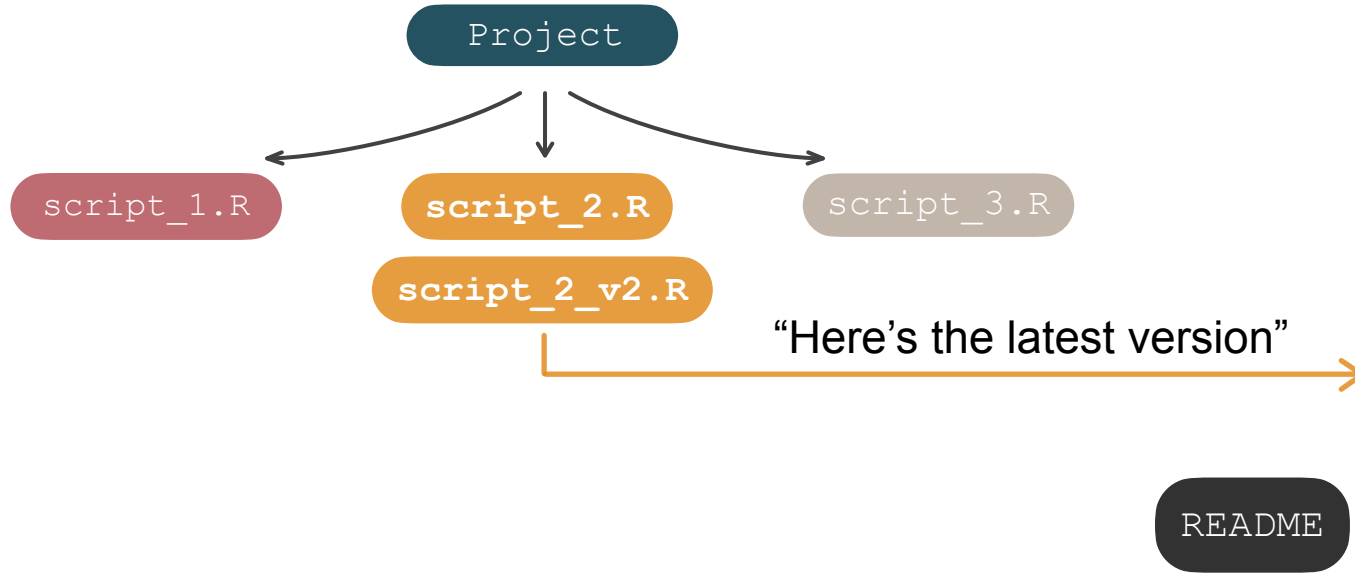
Justification



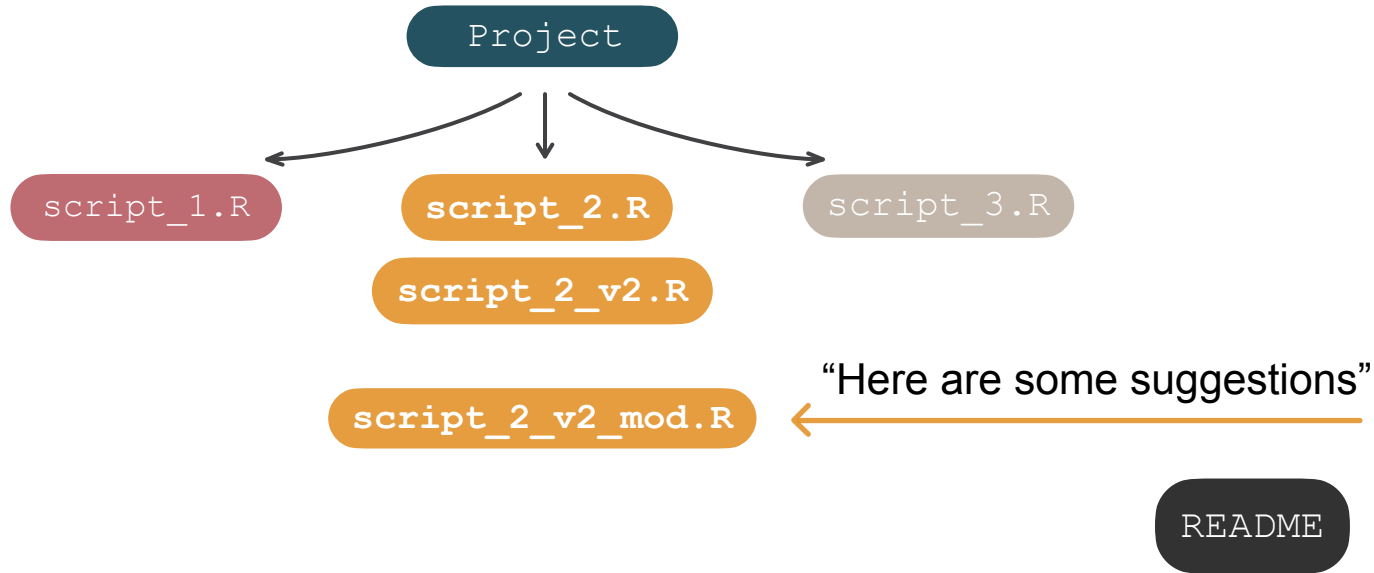
“Can I give it a try?”



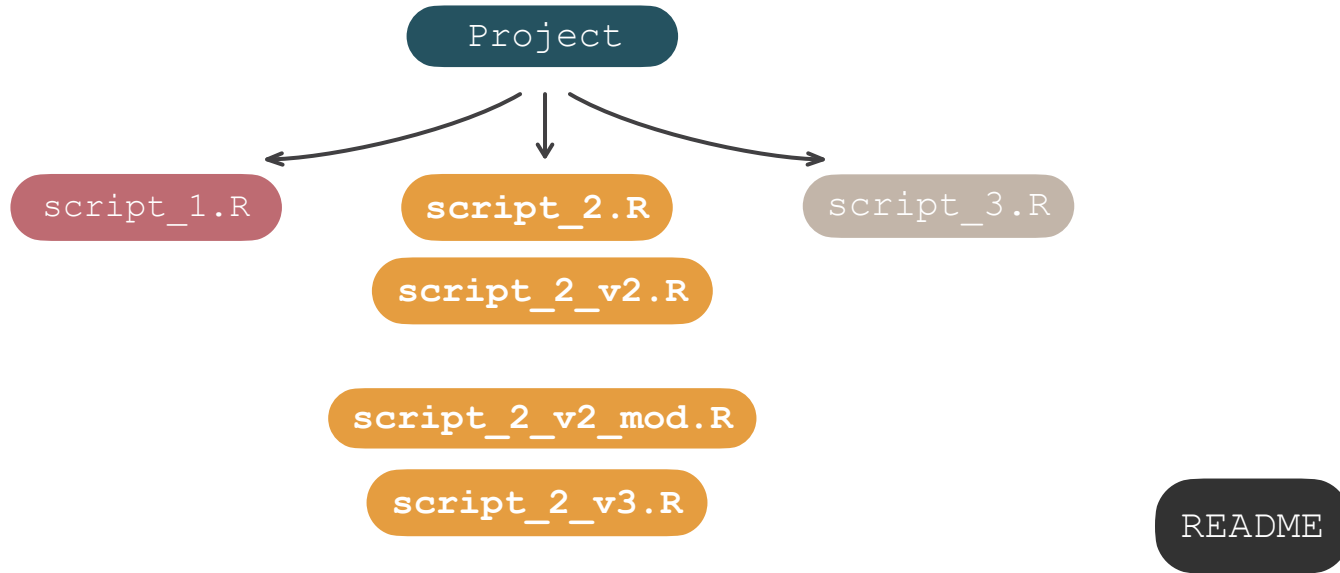
Justification



Justification



Justification



Justification

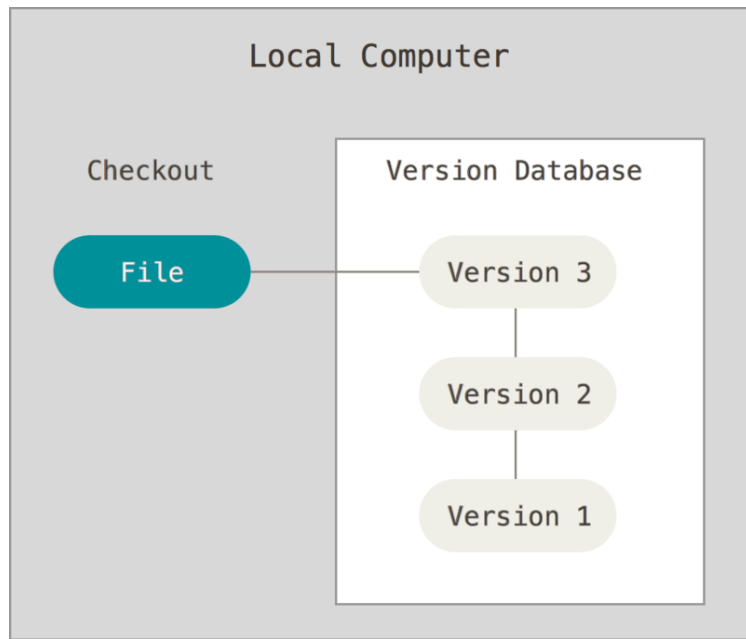
Keeping track of manually means...

- Significant overhead
 - Organization
 - Documentation
 - Communication
- Error-prone and time-consuming
- Makes collaboration difficult
- Six months from now?



Version control systems

"Version control is a system that records changes to a file or set of files over time so that you can recall specific versions later (**history**)"



Version control systems

Referring to and keeping track of versions

- When was “that” change introduced (backtracking)?
- Includes metadata (who, what, when, why)

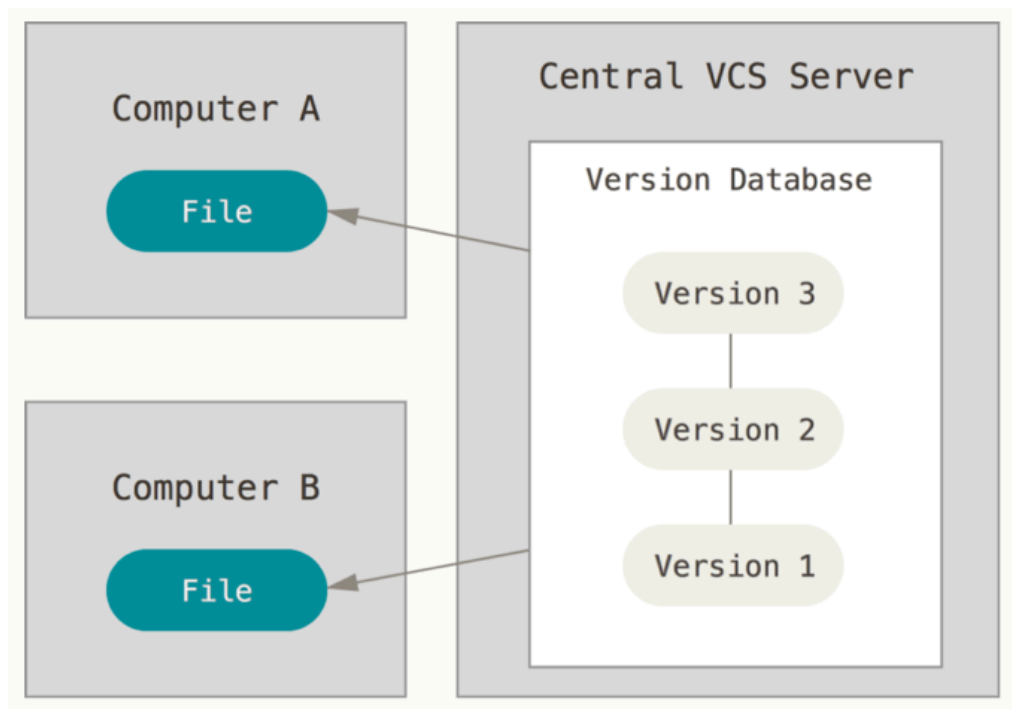
Be organized and maintain long-term control over your code

- Experimenting is easy and organized
- Experiments are self-contained

Keep several versions of code at the same time, without clutter

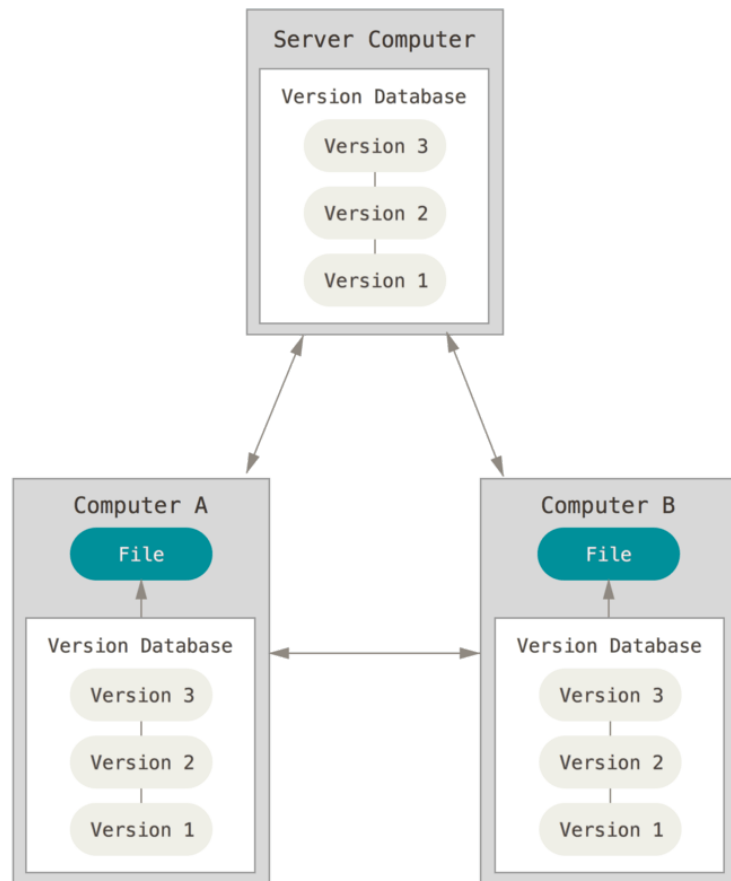
- Keeps known working versions safe
- Able to easy retrieve old versions

Centralized version control systems



Distributed version control systems (DVCS)

- No single point of failure
- Offline work is possible
- Performing most operations is fast
- The system keeps everyone synchronized...
- ...while allowing for asynchronous work



git and GitHub

Git is a DVCS software, created by Linus Torvalds for the management of the Linux kernel.

Other clients exist.

<https://git-scm.com>

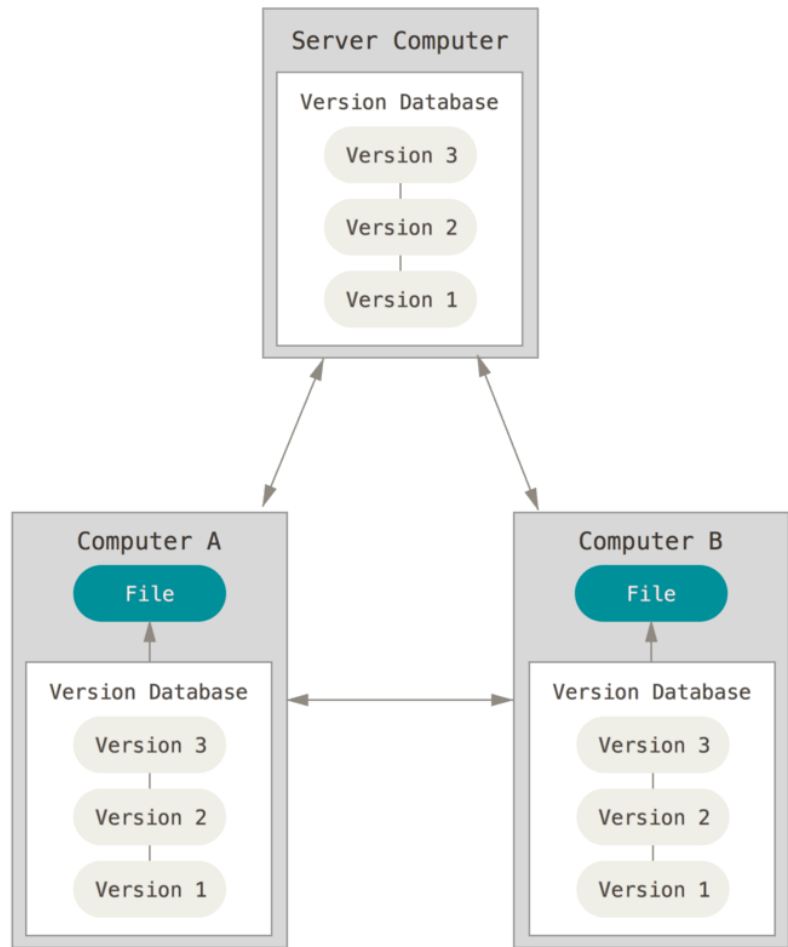


GitHub is a web-based hosting service for version control that uses git

Other services exist.

<https://github.com>

git and GitHub



Useful concepts

Working directory (file/working tree):

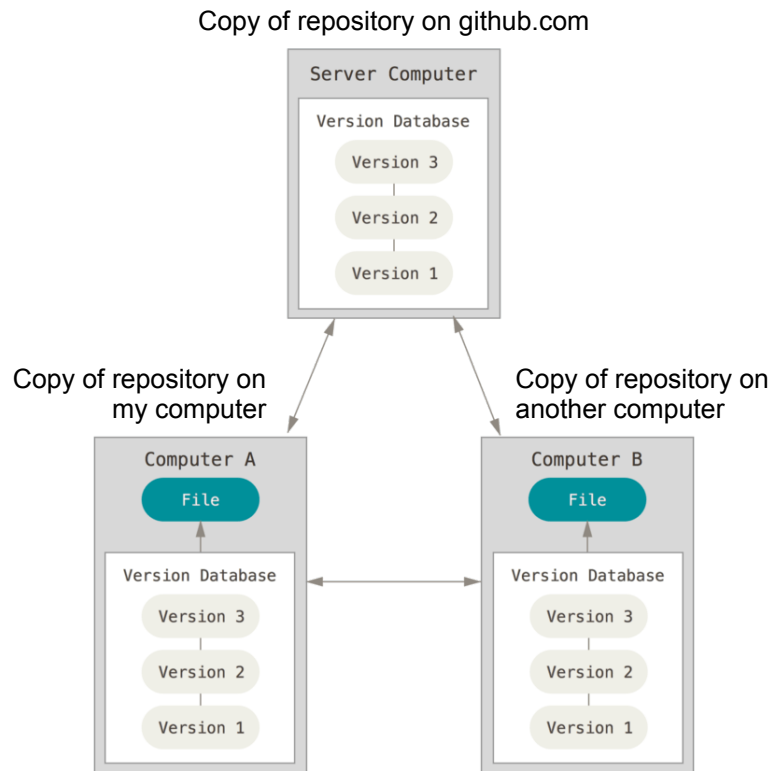
All the files (organized in directories) that we want to keep track of, originating from a single directory.

Repository (repo)

Place in which changes to the working directory are recorded. Can be local or remote. Several copies can exist.

Remote repository

Repository that is linked to our local repository. Often a repository hosted on a site such as GitHub.



Why GitHub?



Helpful web interface: visually browse and explore, community aspect (e.g. issues, wiki), permission control



Public space: visibility, transparency, reproducibility



Easy to *contribute* to projects and track changes



Remote repository (backup)

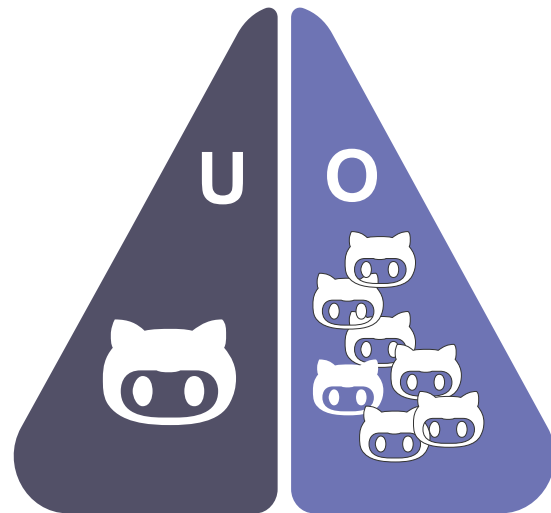


Easy to *collaborate* on the same repository

GitHub

GitHub is organized in

- U** User accounts:
Personal repositories that have a single owner.
- O** Organizations:
 - Designed to better accommodate the needs of a team (e.g. lab).
 - One or multiple owners, and sophisticated access control systems to the repositories.



GitHub

GitHub repositories can be:

PUBLIC

- Accessible to everyone
- Usually released with an open source license
- Still possible to have control over how many people interact with the repository.

PRIVATE

- Not accessible to the public.
- Accessible only from allowed accounts – requires password.

NOTE: Organizations have further options to limit and organize access to repositories

GitHub

	Free \$0/month	Team \$4 per user/month	Enterprise \$21 per user/month	GitHub One
	Join for free	Continue with Team	Contact Sales	Contact Sales
 Code management				
> Public repositories	Unlimited	Unlimited	Unlimited	Unlimited
> Private repositories	Unlimited	Unlimited	Unlimited	Unlimited
 Code workflow				
> GitHub Actions	2,000 minutes/month Free for public repositories	3,000 minutes/month Free for public repositories	50,000 minutes/month Free for public repositories	50,000 minutes/month Free for public repositories
> GitHub Packages	500 MB Free for public repositories	2GB Free for public repositories	50GB Free for public repositories	50GB Free for public repositories
> Code reviews	✓	✓	✓	✓
> Pull requests	✓	✓	✓	✓
> Protected branches	Public repositories	✓	✓	✓
> Code owners	Public repositories	✓	✓	✓
> Draft pull requests	Public repositories	✓	✓	✓
> Multiple pull request assignees	Public repositories	✓	✓	✓

**Academics can get
free “Team” plan!**

A look at git and GitHub

```
git-GitHub-workshop-2022 — -zsh — 101x30
Last login: Mon Feb  7 10:29:21 on ttys000
kgx936@SUN1007442 ~ % cd Desktop/git-GitHub-workshop-2022
kgx936@SUN1007442 git-GitHub-workshop-2022 % ls
ChocolateCake  README.md
kgx936@SUN1007442 git-GitHub-workshop-2022 % git status ChocolateCake/ingredients.txt
```



Search or jump to... Pull requests Issues Codespaces Marketplace Explore

Center-for-Health-Data-Science / **git-GitHub-workshop** Public

Go to file Add file <> Code

January_2023 2 branches 0 tags

"andrejeva-d" Modified the Rstudio part a bit, and added an authentication pop...	f628e58 3 days ago	34 commits
recipes	move example recipes to recipes folder	last week
.DS_Store	Modified the Rstudio part a bit, and added an authentication popup ...	3 days ago
README.md	Update README.md	5 days ago
git-slides_January_2023.key	Modified the Rstudio part a bit, and added an authentication popup ...	3 days ago
git-slides_January_2023.pdf	Modified the Rstudio part a bit, and added an authentication popup ...	3 days ago

README.md

HeaDS git and GitHub workshop 2023

This repository is used for the HeaDS git and GitHub workshop 2023.

The slides for the workshop are available here: [git-slides_January_2023.pdf](#). The [recipes](#) folder contains examples of recipes which can be used for inspiration in the exercises. For the workshops last exercise we will be working with three shared repositories:

About

This repository is used for the HeaDS git and GitHub workshop 2023

Readme 3 stars 5 watching 0 forks

Releases

No releases published
[Create a new release](#)

Packages

No packages published
[Publish your first package](#)

Contributors 4

Let's create a GitHub repo

The screenshot shows the GitHub interface for a repository named "Center-for-Health-Data-Science / git-GitHub-workshop". The repository is public and has 2 branches and 0 tags. The "Code" tab is selected, showing a list of files and folders. A dropdown menu is open in the top right corner, showing options to create a new repository, codespace, gist, or organization. The repository description states it is used for the HeaDS git and GitHub workshop 2023. The README.md file is visible, containing information about the workshop slides and recipes folder.

Search or jump to... Pull requests Issues Codespaces Marketplace Explore

Center-for-Health-Data-Science / git-GitHub-workshop Public

Edit Pins Unwatch 5 Fork 0

<> Code Issues Pull requests Actions Projects Wiki Security Insights Settings

January_2023 2 branches 0 tags Go to file Add file <> Code

"andrejeva-d" Modified the Rstudio part a bit, and added an authentication pop... f628e50 3 days ago 34 commits

File/Folder	Commit Message	Time
recipes	move example recipes to recipes folder	last week
.DS_Store	Modified the Rstudio part a bit, and added an authentication popup ...	3 days ago
README.md	Update README.md	5 days ago
git-slides_January_2023.key	Modified the Rstudio part a bit, and added an authentication popup ...	3 days ago
git-slides_January_2023.pdf	Modified the Rstudio part a bit, and added an authentication popup ...	3 days ago

README.md

HeaDS git and GitHub workshop 2023

This repository is used for the HeaDS git and GitHub workshop 2023.

The slides for the workshop are available here: [git-slides_January_2023.pdf](#). The [recipes](#) folder contains examples of recipes which can be used for inspiration in the exercises. For the workshops last exercise we will be working with three shared repositories:

About

This repository is used for the HeaDS git and GitHub workshop 2023

Readme 3 stars 5 watching 0 forks

Releases

No releases published
[Create a new release](#)

Packages

No packages published
[Publish your first package](#)

Contributors 4

New repository

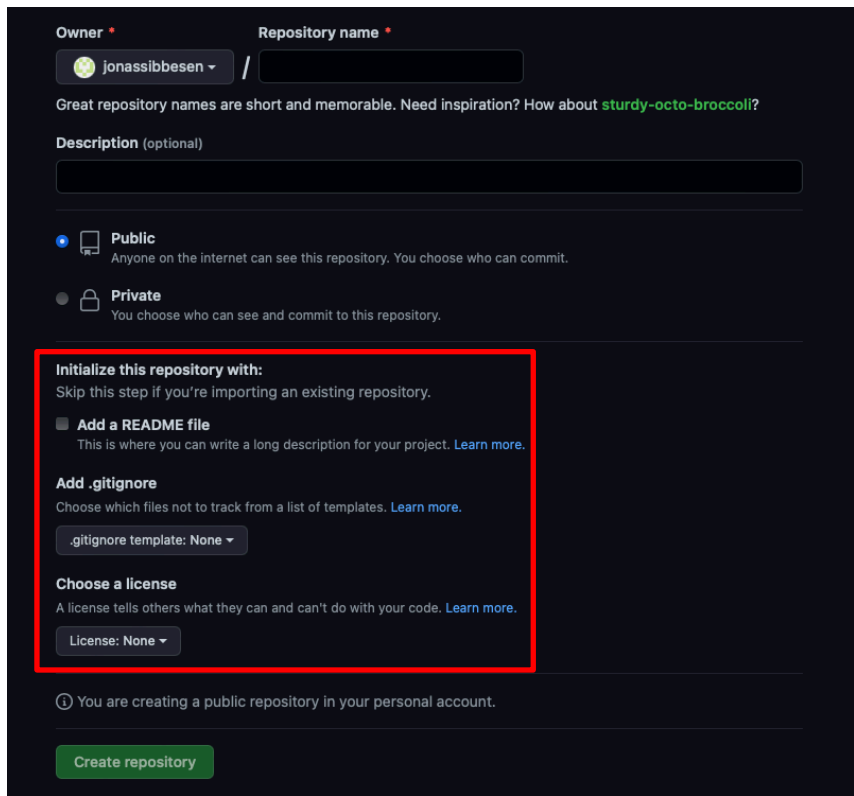
Import repository

New codespace

New gist

New organization

Let's create a GitHub repo



Owner ^{*} Jonassilbesen / Repository name ^{*}

Great repository names are short and memorable. Need inspiration? How about [sturdy-octo-broccoli?](#)

Description (optional)

☒ **Public**
Anyone on the internet can see this repository. You choose who can commit.

☐ **Private**
You choose who can see and commit to this repository.

Initialize this repository with:
Skip this step if you're importing an existing repository.

☒ **Add a README file**
This is where you can write a long description for your project. [Learn more.](#)

Add .gitignore
Choose which files not to track from a list of templates. [Learn more.](#)
.gitignore template: **None**

Choose a license
A license tells others what they can and can't do with your code. [Learn more.](#)
License: **None**

ⁱ You are creating a public repository in your personal account.

Create repository

README file

File shown by default when someone visits the repository – good for an introduction to what the repository is about. Markdown format.

.gitignore

Hidden file containing specifications of files/types that should be ignored at all times by git. There are some default choices.

LICENSE

Creates a LICENSE file that contains the license of the content of your repository.

Choose an open source license

An open source license protects contributors and users. Businesses and savvy developers won't touch a project without this protection.

Which of the following best describes your situation?



I need to work in a community.

Use the **license preferred by the community** you're contributing to or depending on. Your project will fit right in.

If you have a dependency that doesn't have a license, ask its maintainers to **add a license**.



I want it simple and permissive.

The **MIT License** is short and to the point. It lets people do almost anything they want with your project, like making and distributing closed source versions.

Babel, **.NET**, and **Rails** use the MIT License.



I care about sharing improvements.

The **GNU GPLv3** also lets people do almost anything they want with your project, *except* distributing closed source versions.

Ansible, **Bash**, and **GIMP** use the GNU GPLv3.

Commercial:

- MIT License
- Apache License

Open Source:

- GPL v3

git syntax

We will use the terminal command `git`. Syntax:

```
git <command> [<options>] [<args>]
```

Args are usually names, files or directories.

Individual commands have help pages:

```
git <command> --help
```

Examples:

```
git add file.txt
```

```
git commit -m "changes were done"
```

```
git branch -d name
```

Getting started (first time user)

Configure git with your identity:

```
git config --global user.name <username>
git config --global user.email <email>
```

This is important if you want to associate your local changes with your GitHub account. If you want to keep your email private see this guide: <https://docs.github.com/en/account-and-profile/setting-up-and-managing-your-personal-account-on-github/managing-email-preferences/setting-your-commit-email-address>

Change the default text editor used by git (for example to Nano):

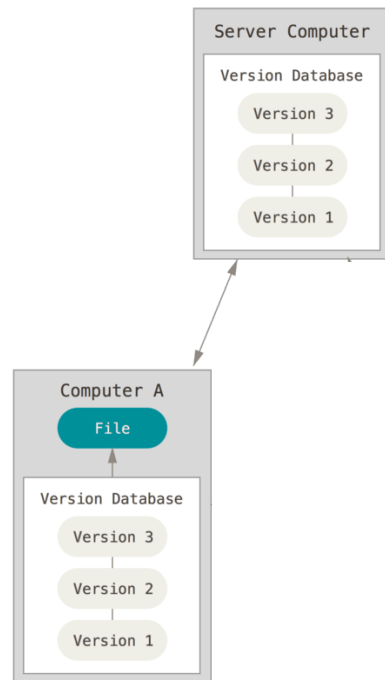
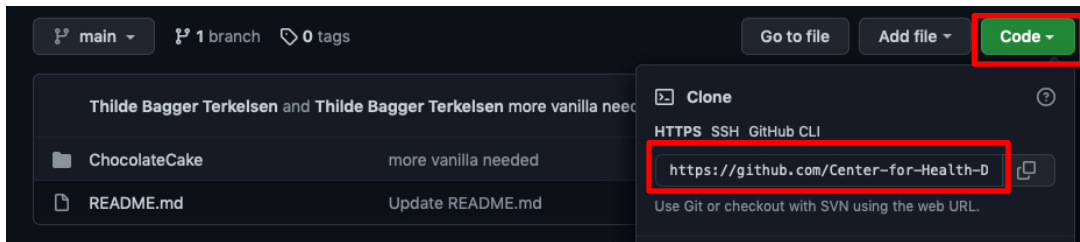
```
git config --global core.editor [text editor]
```

On a Windows system you must specify the full path to the text editor's executable file.

Cloning a GitHub repository

Create a local copy of a remote repository:

```
git clone <GitHub link>  
cd <your repo>
```



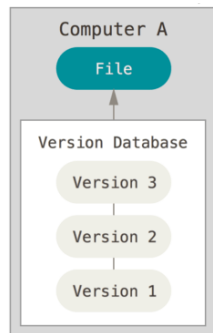
Creating a git repository

It is also possible to create a repository from scratch:

```
mkdir my-repo; cd my-repo  
git init
```

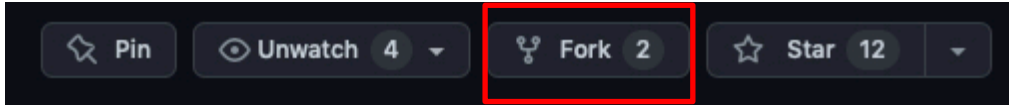
Note that in this case we would not have a remote. Creating a repository on GitHub and cloning it ensures that:

- We have a remote repository
- Remote and local copy are linked



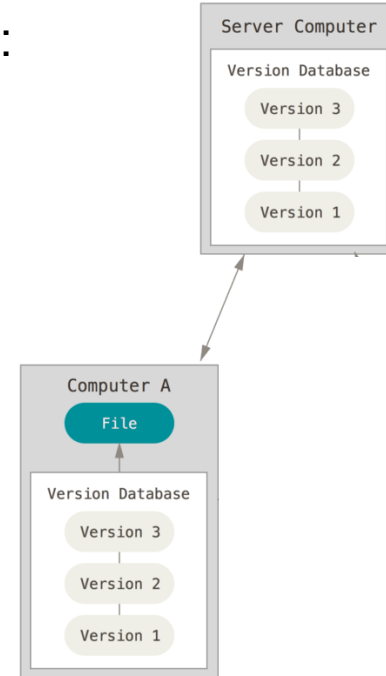
Forking a GitHub repository

Create a copy of someone else's repository on Github:



Forks are most commonly used to:

- Propose changes to another persons repository
- Use another persons repository as a starting point for own project



Checking status of a git repo

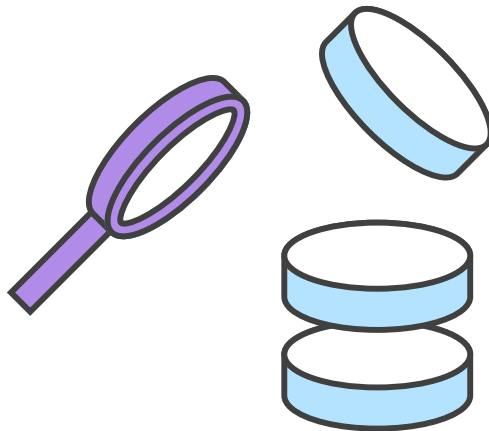
- Check the current state of the repository:

```
git status
```

Gives, among other things, information on which files that have been modified.

- Check the name and location of the remote repository:

```
git remote -v
```



Exercise 1

Exercise

1

- 1. Configure git with username and email if you have not done so before.
- 2. Create a new public repository on GitHub (web interface not on your local terminal). This repo will hold your favourite cooking recipes. Remember to add a README file and choose a license.
- 3. Create a local copy of your GitHub repository by **cloning** it in a folder on your computer. Make note of the location of the directory that you choose to host your local copy.
- 4. Move to the local repository. Check the status of this repository (don't worry if all the information does not make sense yet). What do you think “working tree clean” means? Check the name and location of the remote repository.

Useful concepts

Commit:

Snapshot of a particular state of a working tree. Has an ID & message

Stage:

Selection of changes made on a repository to be added to a commit

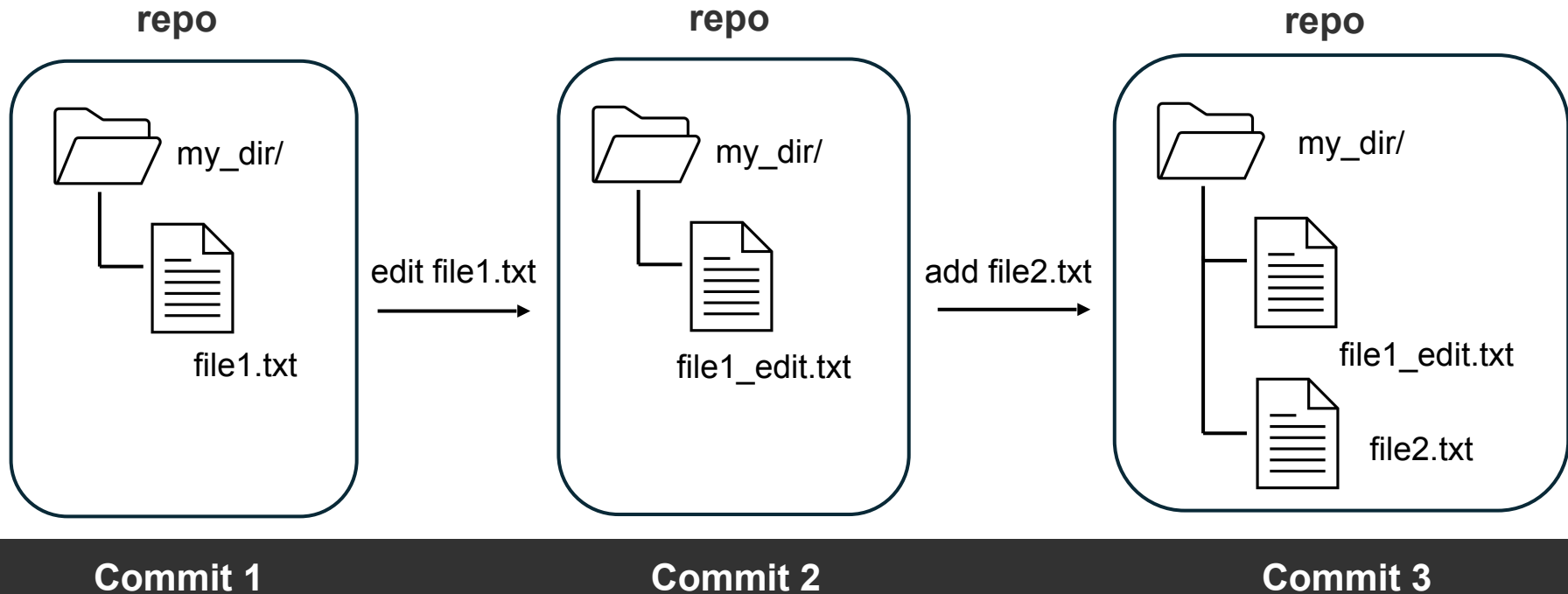
origin = default name of original remote repository

main = default name of main branch (local and remote)

head = current commit (called ***detached head*** when not the tip of a branch)

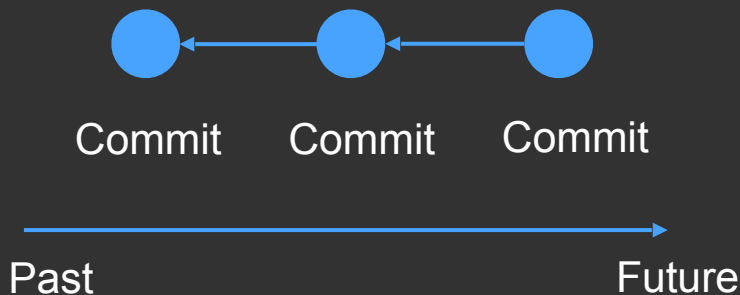
What is a commit?

A commit is a 'snapshot' of the current state of the project:



What is a commit?

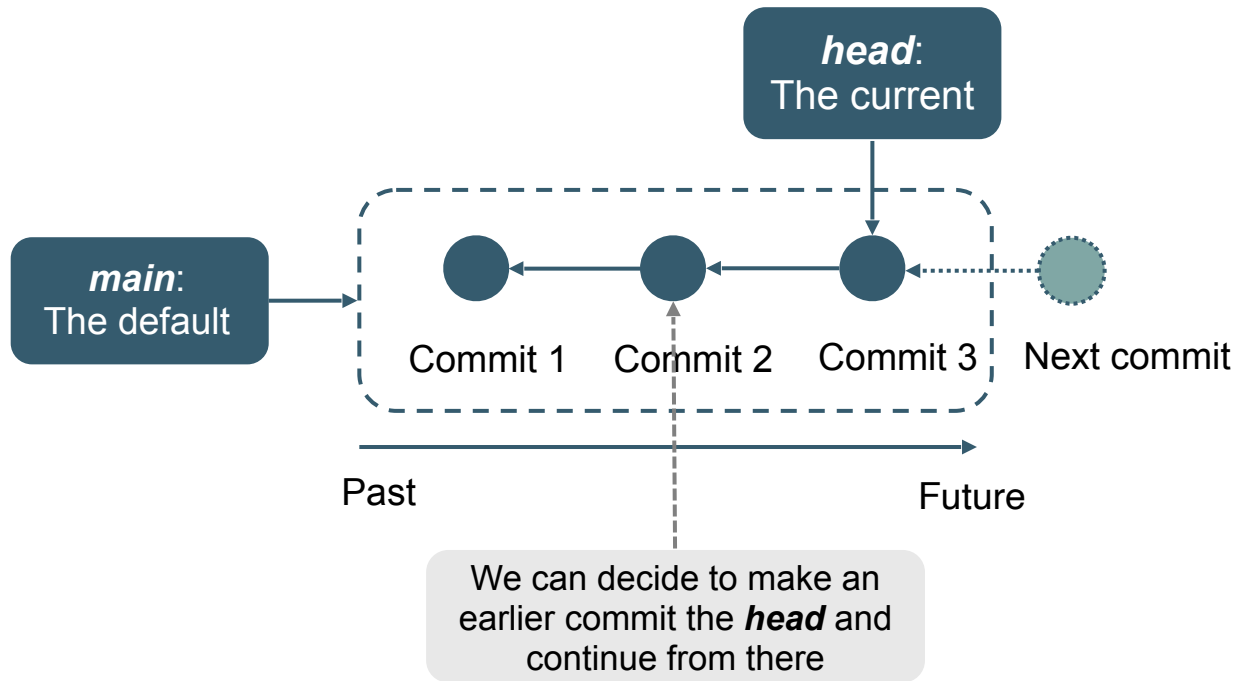
The commit is the atomic unit of a git working tree.
They are often illustrated with a circle in schematics:



When we show the
commit history, we will
often paint arrows
pointing backwards

Commits do not store the actual files but the current state of the repo

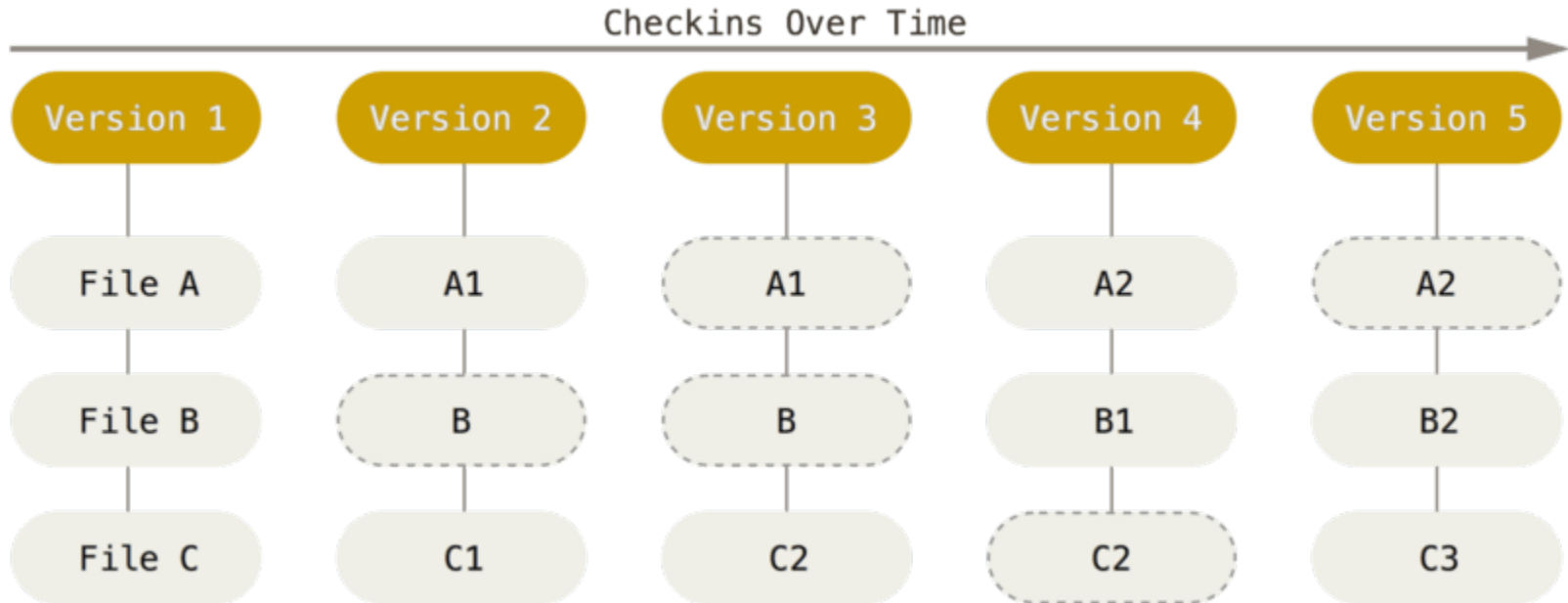
Useful concepts



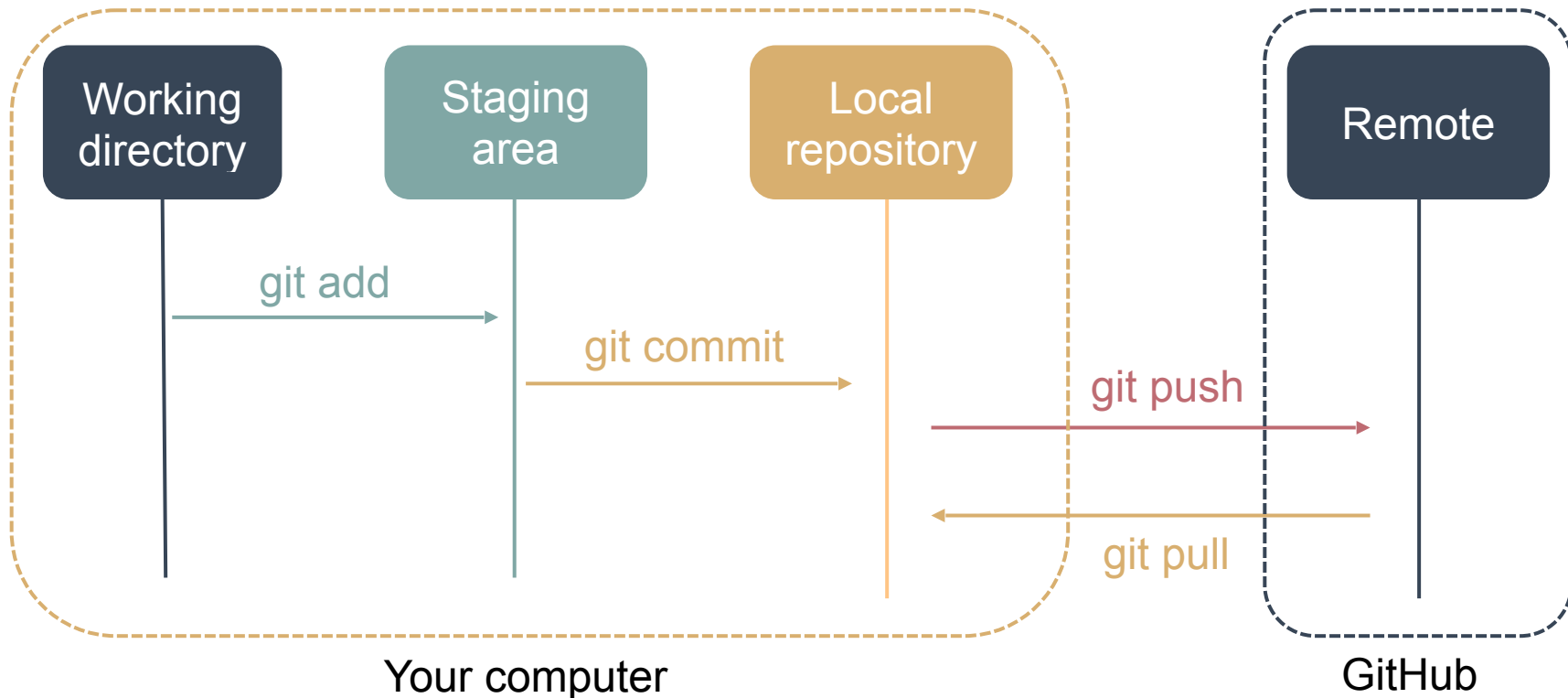
New commits are built on / point back to the last **head**

Commits have:
ID (=hash)
Message

Git stores snapshots of the files



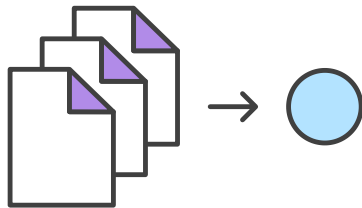
What is staging



Adding files to stage

Files are added to stage using the command **add**

```
git add <filename>
```



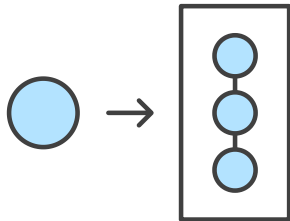
- One can stage multiple files or even directories (if they contain files)
- Option `-A` stages all changes if no arguments are given
- More than one filename are allowed including wildcards

To remove or move a file (including staging the change) use:

```
git rm <filename>
```

```
git mv <filename>
```


When you're happy with your changes and have added all the files to stage the next step is to commit the changes:



```
git commit
```

```
git commit -m "Commit Message"
```

Use `-a` to automatically stage files that have been modified or deleted. **New files you have not told git about are not affected.**

```
git commit -a -m "Commit Message"
```

First Commit

Good practices

- **Commits** should be:
 - Atomic and self-contained (or WIP), but not excessively pedantic
 - Done often (try not to have gigantic blob commits)
- **Commit messages** should:
 - Always have a short comment (more descriptive than “bugfix”, “save”, etc.)
 - Written to be understandable by someone else >1 year down the line, including you

Checking history

Check the history of a repository:

```
git log
```

A few useful options:

- To see only the commits of a certain author

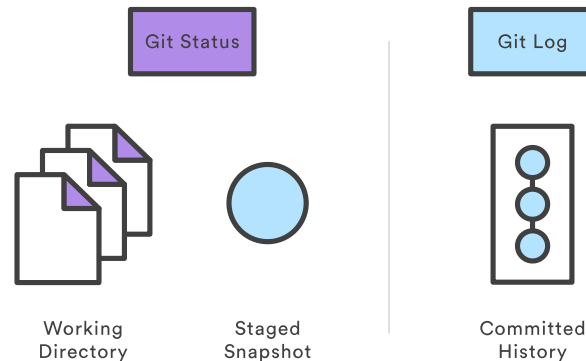
```
git log --author=<name>
```

- To see a compact history where each commit is only one line

```
git log --oneline
```

- To see the history as a graph (useful when having multiple branches):

```
git log --all --decorate --oneline --graph
```



Checking differences

Check

- not staged changes since the last commit:

```
git diff
```

- staged changes since the last commit:

```
git diff --cached
```

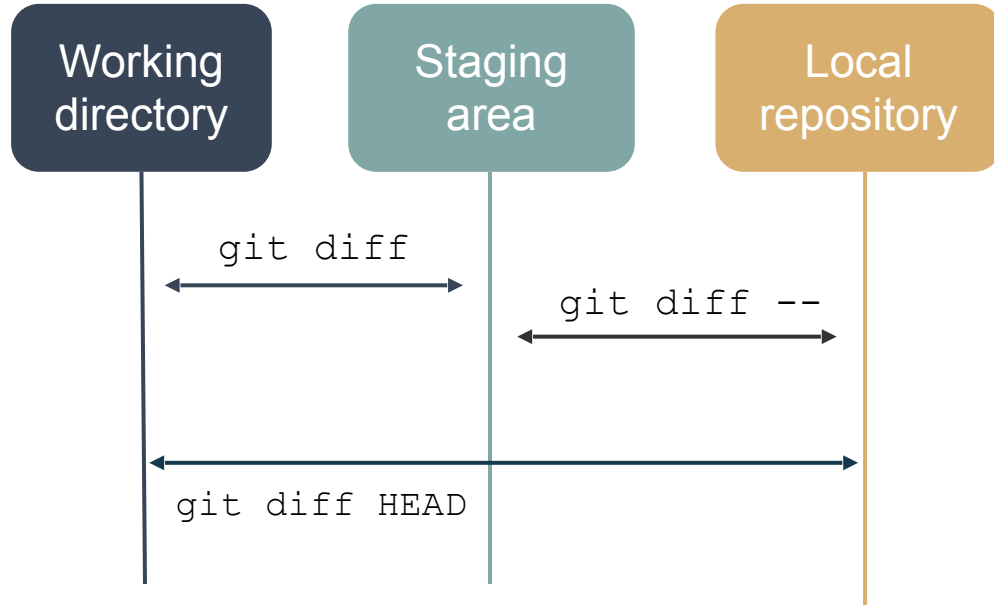
- differences between two commits:

```
git diff <ID1> <ID2>
```

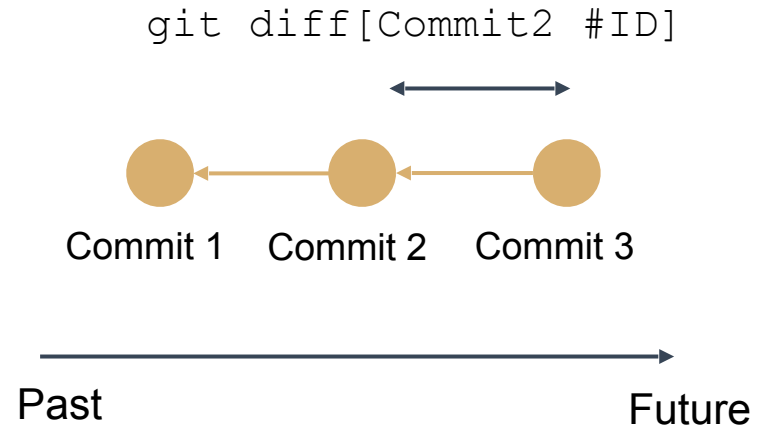
- differences between second last and last commit:

```
git diff HEAD^ HEAD
```

Checking differences



Differences between



Differences between

Exercise 2

Exercise

2

- 1. Create two files named `ingredients.txt` and `recipe.txt` (can contain anything you would like) in the repository folder on your local computer.
- 2. Check the status of the repository. **Stage** the new files (use `git add`) and check the status. **Commit** the staged files and check the status. What did you see change between each step when checking status?
- 3. Create a directory with the recipe name and move your files to it with a git command. Then create a commit with your changes.
- 4. Add or remove one step in *recipe.txt* and check the change before committing (use `git diff`). Stage and commit your change.

Exercise 2 cont.

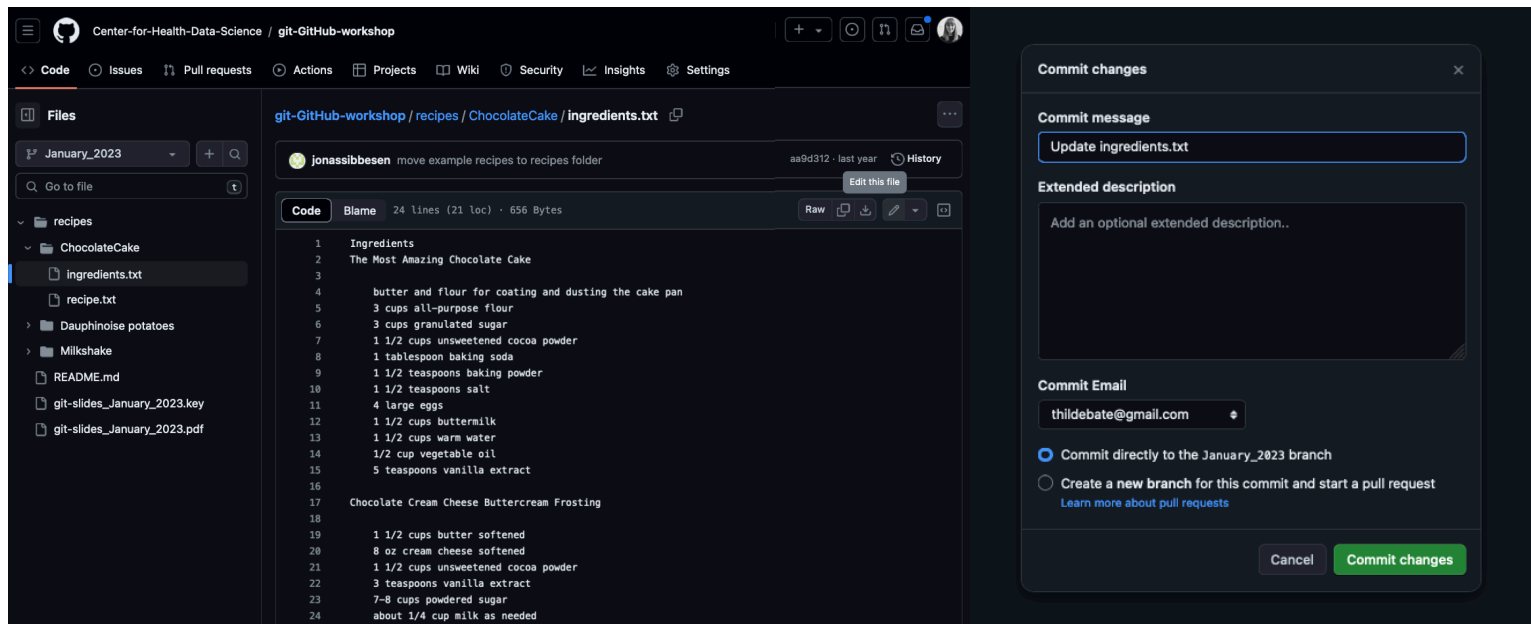
Exercise

2

- 5. Check the commit history. Can you identify the author, ID, time and date of your commits? Try experimenting with the different ways of showing the history.
- 6. Check the difference between different commits. Can you identify the filename(s) and changes?

GitHub Intermezzo

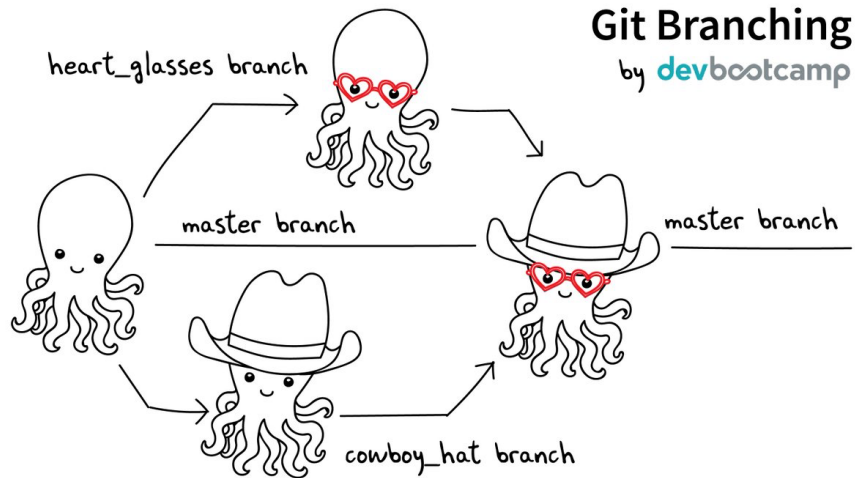
- You can edit files and commit changes directly from GitHub
- Not ideal when working with collaborators as it is more likely to create conflicts when local versions are merged/pushed



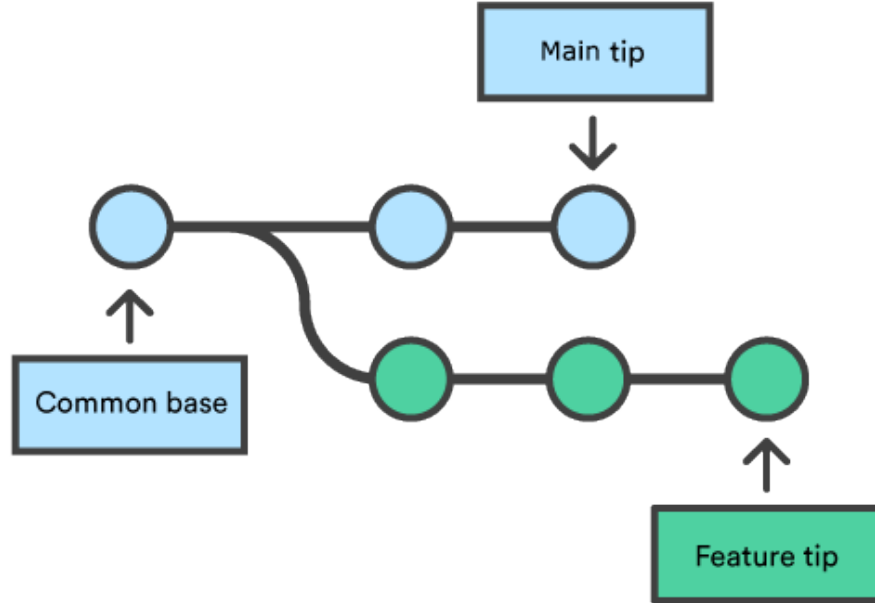
Branching

A **branch** represents an independent line/track of work on your repository.

- Keep a branch that is in working order (*main* branch).
- Isolate experiments from the *main* branch.
- Merge changes we like into the *main* branch.
- Work on more than one feature at the same time, independently.



Branching



- List all the branches in your repository:

```
git branch -a
```

- Create a new branch:

```
git branch <branch>
```

- Switch to a branch:

```
git checkout <branch>
```

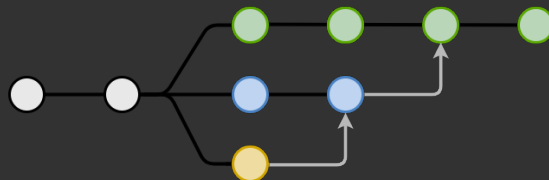
- Safe deletion: prevents deleting if branch has unmerged changes:

```
git branch -d <branch>
```

- Force delete:

```
git branch -D <branch>
```

Branching



Good practices

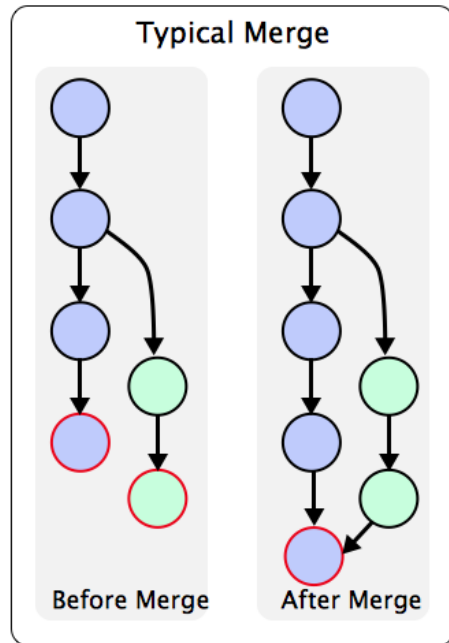
- **Branches** should:
 - Have descriptive names.
 - Done every time you want to experiment (e.g. add a new feature or fix a bug).
 - Have a single purpose (e.g. one feature).
 - Derived from a sensible commit (often the latest commit on the *main* branch)

Merging

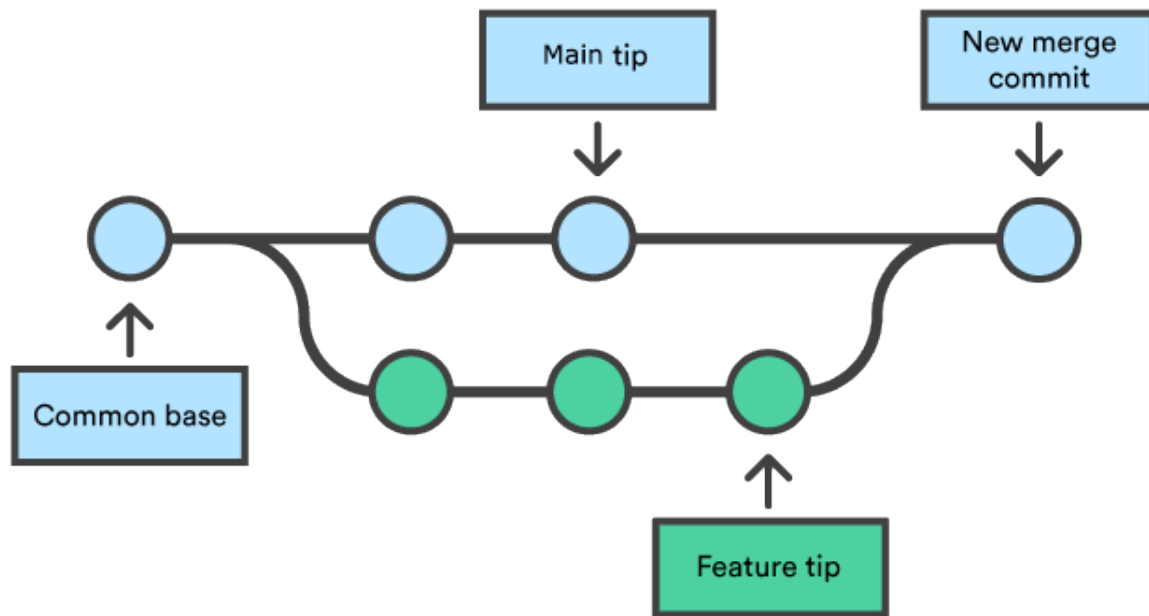
Merging incorporates changes made in another branch **into the current/active branch.**

- **Switch** to the branch you want to change (not the one you want to merge)
- **Merge** changes made in another branch **into the current/active branch.**

```
git checkout <active branch>  
git merge <branch to merge>
```



Merging



Merging

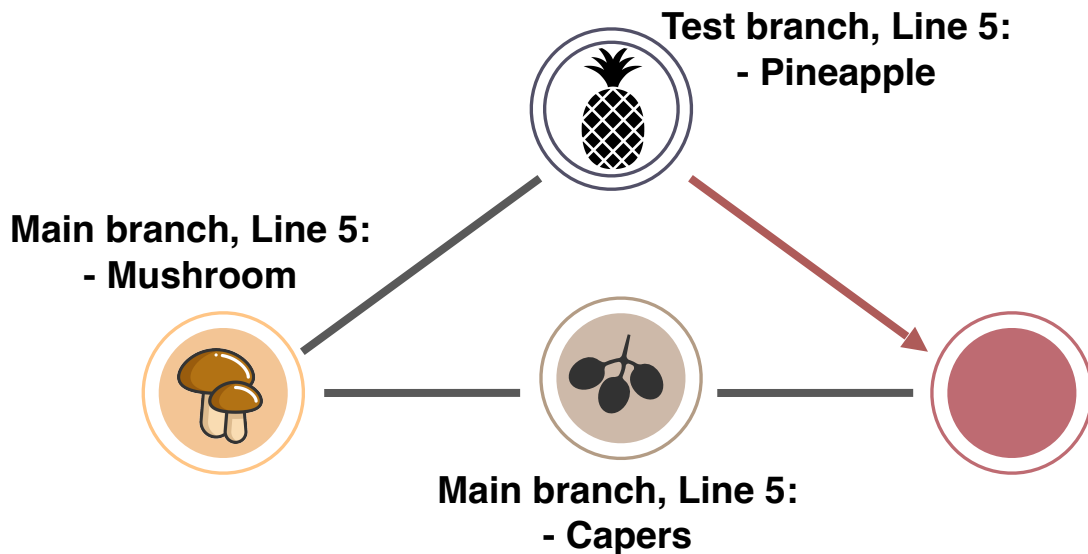
Two main cases for merging

We want to integrate a feature we were working on in a branch into *main*.

main has diverged enough from our branch that we want to integrate the new commits in it before merging our feature – so we merge *main* into our feature branch.

Conflicts

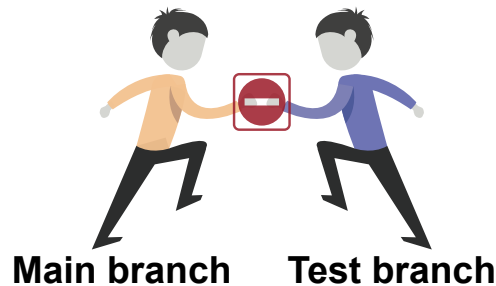
A merge conflict happens when trying to merge branches that have changes **in the same files and in the same lines**



Conflicts

```
(base) Matteos-MacBook-Pro-2:test matteotiberti$ git merge test
Auto-merging recipe.txt
CONFLICT (content): Merge conflict in recipe.txt
Automatic merge failed; fix conflicts and then commit the result.
```

- Check which file(s) have conflicts.
- Modify files by hand to achieve the desired result.
- Add all files to stage once they are ready and commit. This solves the merge in a commit.
- **Remember to remove all conflict markers** and change your code in the final version you want.



```
- tomato
- mozzarella
- flour
- water
<<<<<<< HEAD
- capers
=====
- pineapple
>>>>>>> test
```

Exercise 3

Exercise

3

- 1. Create an ***experimental*** branch and switch to it. Change an ingredient in *ingredients.txt* and store the change as a commit. Try to get a list of the branches in the repository.
- 2. Go back to your ***main*** branch and add a step in *recipes.txt* and store the change as a commit. Check the difference between the two branches (use `git diff`).
- 3. Merge your ***experimental*** branch into your ***main*** branch. Check the history (try using the graph output).

Exercise 3 cont.

Exercise

3

- 4. Switch to your ***experimental*** branch and change the first line in one of your files, commit your changes.
- 5. Switch to your ***main*** branch and change the same first line in the same file in a different way, and commit.
- 6. Try to merge your ***experimental*** branch into your ***main*** branch; you should get a conflict. Check the status of the repository.
- 7. Edit the conflicting file to solve the conflict. Stage and commit to fix it. Check the history (try using the graph output).

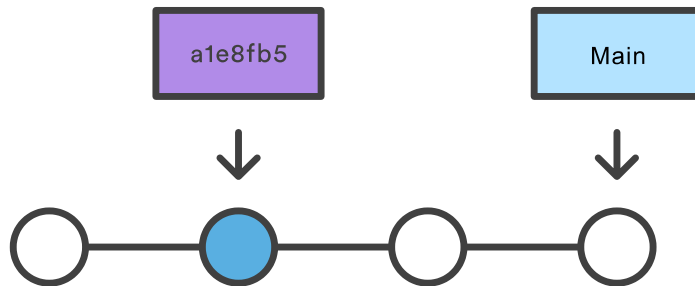
Back to the past

We can move around the repository using the command **checkout**

```
git checkout <identifier>
```

Changes the local content of the working directory to that of the reference - which means, moves **HEAD** to identifier. Identifier can be:

- A commit **ID**
- A branch **name**



Back to the past

Rewriting history is (usually) a bad idea

Every change you do in git is done with a forward in time perspective - the past should generally not be changed.

- Changes (even to something done a long time ago) should be done in commits added to the HEAD
- To restart from an earlier commit use branching/merging

I've botched staging or commit!

How to fix the last commit message:

```
git commit --amend -m "new message"
```

- Do not amend a message of a commit that have already been shared with others (e.g. uploaded to GitHub).



I've botched staging or commit!

`reset` is useful to “undo” some of the changes:

```
git reset
```

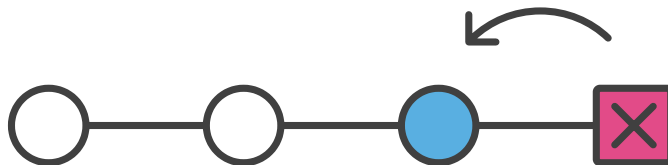
- Removes all staged changes.

```
git reset <filename>
```

- Removes file from staged area.

```
git reset --hard <ID>
```

- Moves reference to that commit (can be HEAD) and resets the working directory to match that commit. **Destructive operation** - only on local branches that have not been shared with others yet (**do not use on *main***).



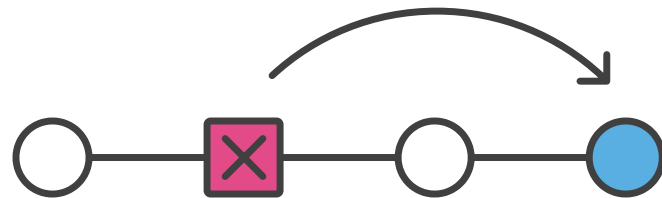
I've botched staging or commit!

revert is used to add commits that “reverse” the changes done by older commits:

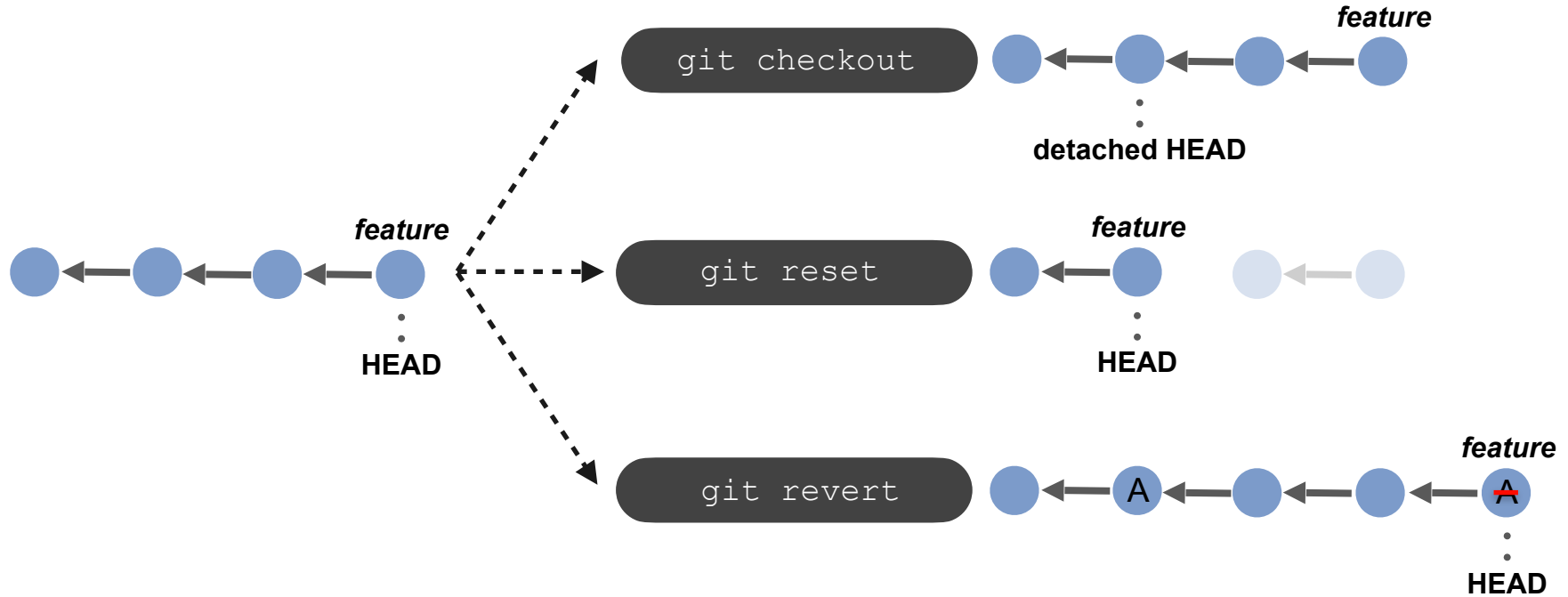
```
git revert <ID>
```

```
git revert <ID1>..<ID2>
```

- Here “ID” is the commit that you want to revert
- Non-destructive operation (history is maintained)
- History is polluted with commits that just revert – not the cleanest option
- Better than reset for shared repositories



Undoing changes



Exercise 4

Exercise

4

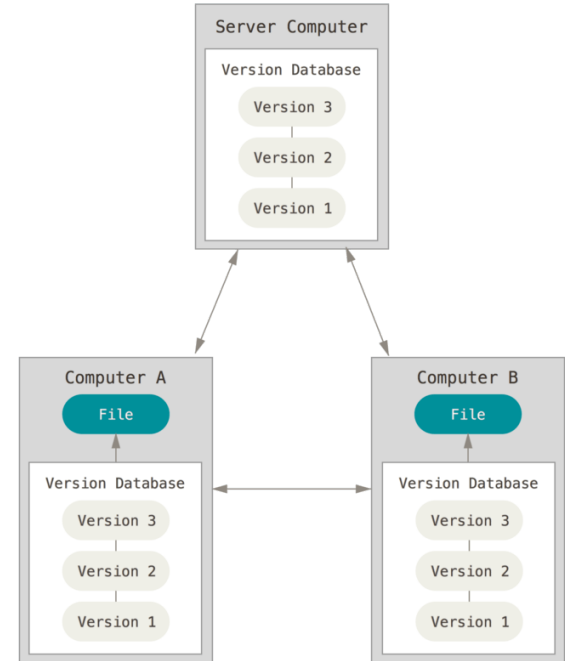
- 1. Checkout an earlier commit on the **main** branch. Check the content of the files. Check the history and status of the repository. What do you notice? Checkout **main** to get back to the tip of the branch.
- 2. Switch to the **experimental** branch. Add a step in *recipes.txt* and add the changes to stage, but **do not commit**. Check the status. Do a hard reset of **HEAD** and check the files and status. What do you notice?
- 3. Add a new ingredient to *ingredients.txt* and commit the changes. Add a new step to *recipes.txt* and commit the changes. Change the commit message of the last commit and check the history to see the change.
- 4. **Revert** the first of the two new commits (it will ask you to write a commit message). Check the files and history. What do you notice? Switch back to **main**.

Interacting with a remote

A **remote** is a repository that is linked to our local (here our local repository started out as a copy of a remote repository on GitHub).

Changes (commits) in the local repository can be uploaded and merged into the remote repository.

Changes that have happened in the remote repository (e.g. other people's) can be downloaded and merged into the local repository.



Personal Access Token (PAT)

<https://docs.github.com/en/authentication/keeping-your-account-and-data-secure/managing-your-personal-access-tokens>

Personal access token to use with the command line.

The token works as a password to the remote GitHub repository.



Advantage of PAT over password:

- **Unique:** Can be generated per use or per device.
- **Revokable:** Can revoke access to each one at any time.
- **Secure:** Random strings of characters that cannot be attacked by brute force i.e. less hackable than a password.
- **Quantity:** Any number of access tokens can be created, compared to only one password per user.

Interacting with a remote

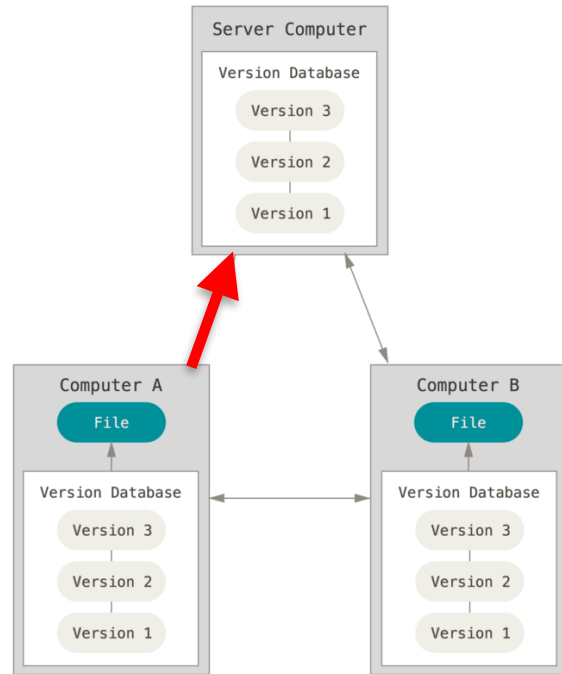
- Use **push** to upload local repository content to remote (i.e. export commits to remote branches)

```
git push <remote> <branch>
```

If this is your first time pushing to GitHub you need to login with your username and PAT.

- The default remote branch for the current local branch can be set when pushing (set upstream):

```
git push -u <remote> <branch>
```



Interacting with a remote - Windows users

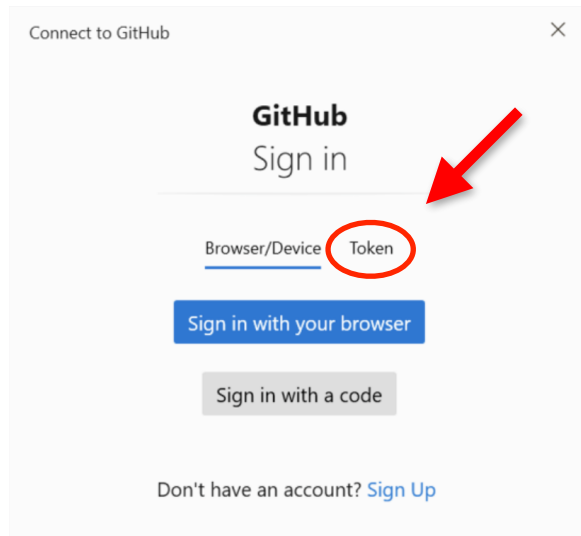
- Use `push` to upload local repository content to remote (i.e. export commits to remote branches)

```
git push <remote> <branch>
```

If this is your first time pushing to GitHub you need to login with your username and PAT.

- The default remote branch for the current local branch can be set when pushing (set upstream):

```
git push -u <remote> <branch>
```



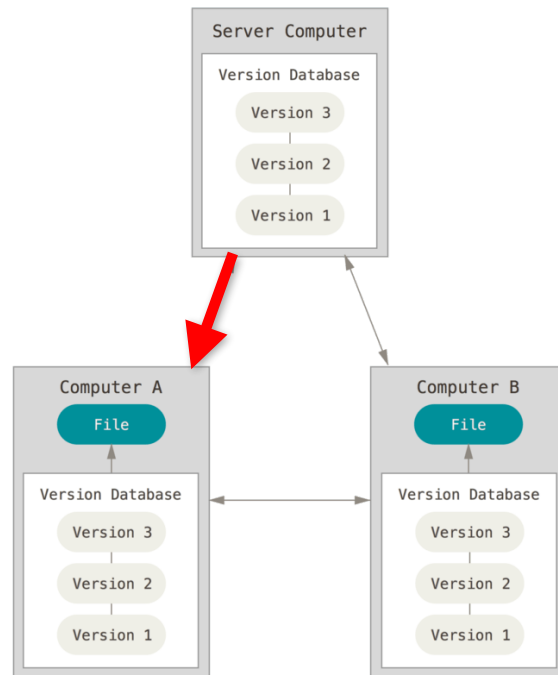
Interacting with a remote

- Use **pull** to download and apply the changes that were done in remote to your local repository (i.e. apply commits by other people):

```
git pull <remote> <branch>
```

- Use **fetch** to download without merging (remote fetched branches are named as “<remote>/<branch>”):

```
git fetch <remote>
```



Exercise 5

Exercise

5

- 1. Go to your GitHub account and set up your personal access token (if you have not already done it). Follow the **Creating a personal access token (classic)** step-by-step on GitHub.com (click **repo** when selecting scopes):
<https://docs.github.com/en/authentication/keeping-your-account-and-data-secure/managing-your-personal-access-tokens#creating-a-personal-access-token-classic>
- 2. Copy your access token and note your GitHub username as you will need both the first time you are pushing data to GitHub and for the RStudio part. If you have pushed to GitHub before you might not need to use it.

Exercise 5 cont.

Exercise

5

- 3. Push your changes to **your** remote repository for your **main** branch. Visit your remote repository on github.com. Check if your changes are there.
- 4. Push the **experimental** branch to your repository. Check the changes on GitHub. What do you notice?
- 5. Merge the **experimental** branch into **main**. Push your changes and check them on GitHub. What has changed?

Exercise 5 cont.

Exercise

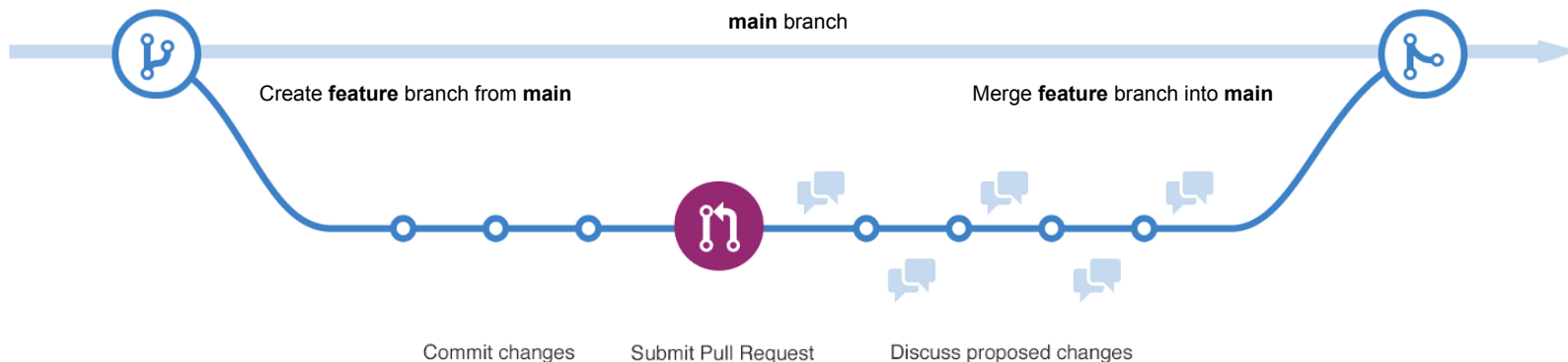
5

- 6. Edit the **README.md file on GitHub**. Create a new commit on **main** and try to push it. What happens?
- 7. **Fetch** the changes from the remote repository. Note that the remote version of the **main** branch is named **origin/main**. Look at the difference between the local and remote version of **main**.
- 8. Merge remote version **origin/main** into the local version of **main** and push the changes to the remote repository on GitHub. Visit GitHub and check if your changes are there.

Working together on GitHub

Contributing and interacting with other users

- Projects on GitHub often follow a pattern of development and communication.
- The *main* branch is considered the “stable” branch and is not changeable directly - only through merges and pull requests.
- Contributions happen in branches that are merged into *main*.
- Releases happens once in a while from the *main* branch.



Working together on GitHub

Interacting with other users: README



Read the documentation of the project you want to contribute to or use.

README.md



git-GitHub-workshop-2023

This repository is used for the HeaDS git and GitHub workshop 2023. Workshop participants should clone the repo after which they can use it to complete workshop exercises associated. The shared repo is made for hosting cooking recipies which are made locally by participants and then pushed to remote for everyone to view.

HeaDS workshop contacts:

Working together on GitHub

Interacting with other users: Issues



Check *Issues* if you have a problem or question to see if others have already asked the same. If needed, open a new issue.

The screenshot shows the GitHub interface for the repository **ELELAB / mutatex**. At the top, there are buttons for Unwatch (2), Star (7), and Fork (2). Below this is a navigation bar with links to Code, Issues (11), Pull requests, Actions, Projects, Wiki, Security, Insights, and Settings. The **Issues** tab is selected and underlined.

A message box states: "Label issues and pull requests for new contributors. Now, GitHub will help potential first-time contributors discover issues labeled with **good first issue**." There is a "Go to Labels" button and a "Dismiss" link.

Below the message is a search bar with the filter "Is: issue is: open". To the right are buttons for "Labels 8" and "Milestones 1", and a green "New issue" button.

The main content area displays a list of issues. Each issue has a checkbox, a status icon (green circle with a checkmark), a title, a label (blue pill with "enhancement"), and a comment icon (speech bubble). The issues listed are:

- ☐ **Wild type residues types should be checked when using a position list** **enhancement**
#98 opened 7 days ago by mtiberti
- ☐ **Mutation runs are getting terminated before completing**
#95 opened on 31 Jul by asirajee
- ☐ **Add individual data points in histogram and box plots**
#93 opened on 2 Apr by emilianomaiani
- ☐ **Unrecognised residues from FoldX** **enhancement**
#92 opened on 13 Feb by juansv

Working together on GitHub

Interacting with other users: Branches



Create a local copy of the repository.

- **Clone** if you have write access to the repository (direct access as *collaborator* or part of the repo *organization*). Or if you are only planning on using the method and not contributing to its development.
- **Fork** if you have no affiliation to the repository or you want your own copy to experiment with.



Create a **new branch** from the *main* branch. Add changes as commits. Push them so they are present on the remote (GitHub).

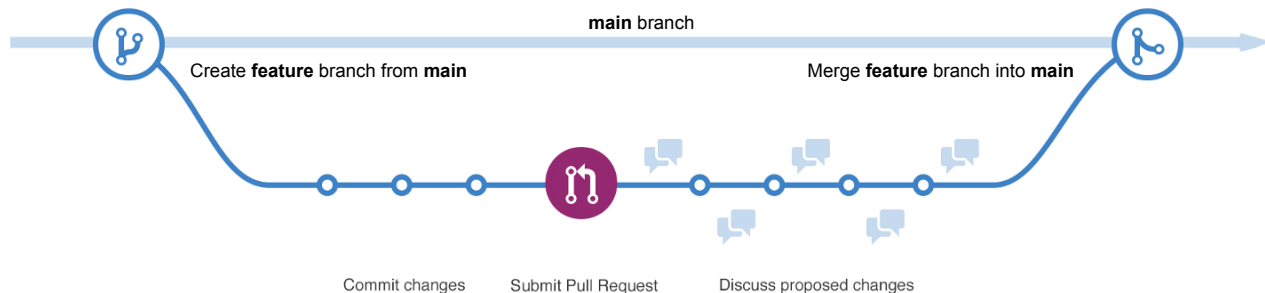
Working together on GitHub

Interacting with other users: Pull requests (PRs)



Create a *Pull Request* (*merge request*) from the new branch.

- Your code might be subject to code review. If so, the maintainers will decide if/when your code is good enough to be merged.
- To update the PR push the new changes to the same branch.
- Link to related issues in the PR. Use keywords (e.g. “resolves”) to automatically close an issue when the PR is merged into *main*: “This PR resolves issue #40”.



Working together on GitHub

Interacting with other users: Pull requests (PRs)

The screenshot shows a GitHub Pull Request (PR) interface. At the top, there's a navigation bar with tabs for Issues (2), Pull requests (selected), Actions, Projects, Security, and Insights. The PR title is "Update Catch2 to newer version #41" with a green "New issue" button to the right. Below the title, a purple "Merged" badge indicates that "jonassibbesen merged 1 commit into master from update-deps 3 weeks ago". A summary bar shows "Conversation 0", "Commits 1", "Checks 0", and "Files changed 1", along with a diff indicator "+1 -1". The main content area shows a comment from "jonassibbesen" (owner) stating: "Updated the Catch2 submodule to a more recent version. Please run `git submodule update --init --recursive` to update the submodules before compiling. This resolves #40". Below the comment is a commit card for "Update Catch2 to newer version" (commit 928efb8). Further down, a merge card shows "jonassibbesen merged commit d0478d0 into master 3 weeks ago". At the bottom, a card indicates "jonassibbesen deleted the update-deps branch 3 weeks ago". On the right side, there are sections for "Reviewers" (No reviews), "Assignees" (No one assigned), "Labels" (None yet), and "Projects" (None yet).

Exercise 6

Exercise

6

- 1. Go to the shared recipes repository in GitHub for which you have become a collaborator.
- 2. Open an **issue** about a recipe you think is missing.
- 3. Open your terminal, make a local copy of the shared repository (where you just opened an issue) with **clone**.
- 4. Create a new branch. Switch to this branch.
- 5. Add your recipe to the new branch. Commit and push your changes (HINT: for pushing correctly recall what you did in exercise 5).

Exercise 6 cont.

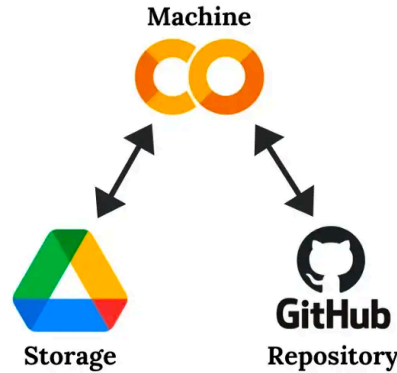
Exercise

6

- 6. Go back to the GitHub repository (in the browser) and open a **Pull request (compare and pull)** with your new branch.
- 7. In your **pull request** message link to your **issue** with one of the **keywords** listed [here](#)
- 8. Wait for one of us to approve the PR. Pull from the repository and check our shared recipe book.

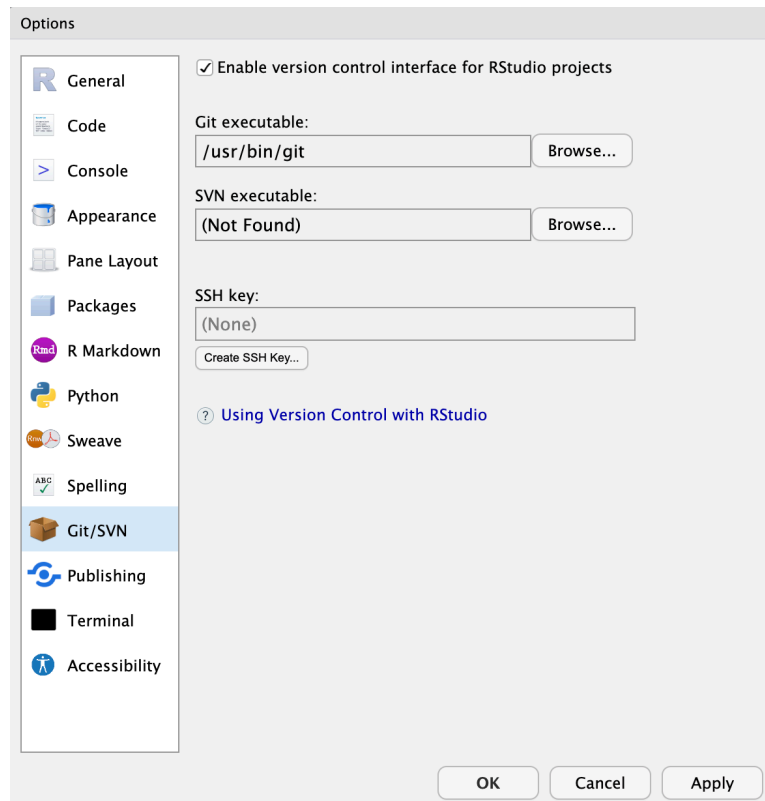
Git & GitHub with Code Editors

- You can work with **git/GitHub** from within code-editors, including Rstudio (R), PyCharm, Visual Studio etc. (Python).

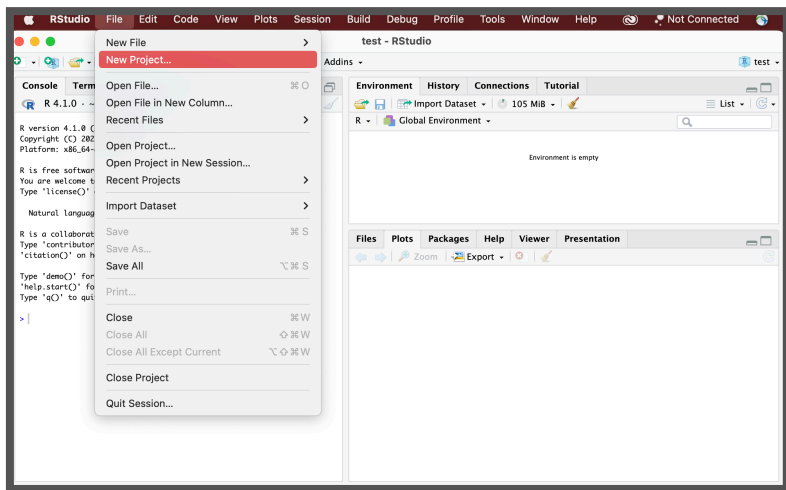


Configure RStudio

- **Make Sure RStudio Knows about Git**
- Go to the Edit → Settings → Git/SVN
- Enable the version control interface and make sure RStudio knows where to find git.



Git & RStudio



Researchers working with R benefit from using version control.

From within RStudio Project!

Notice where the Project will be saved locally!



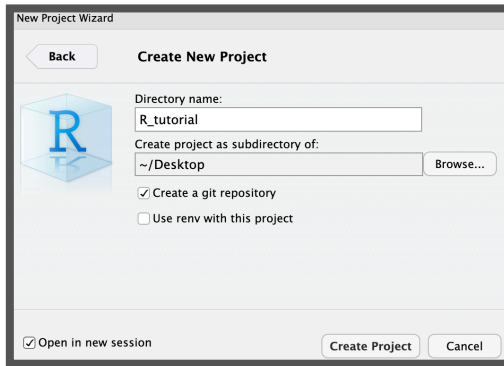
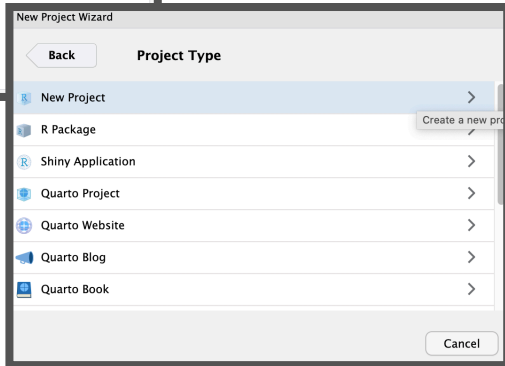
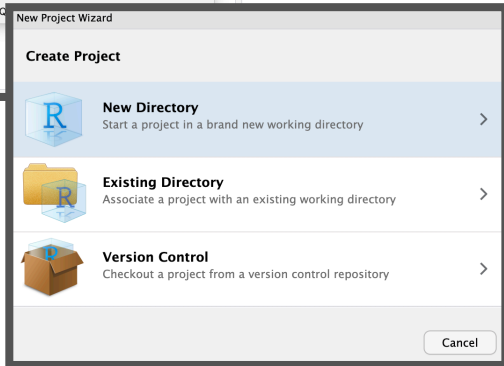
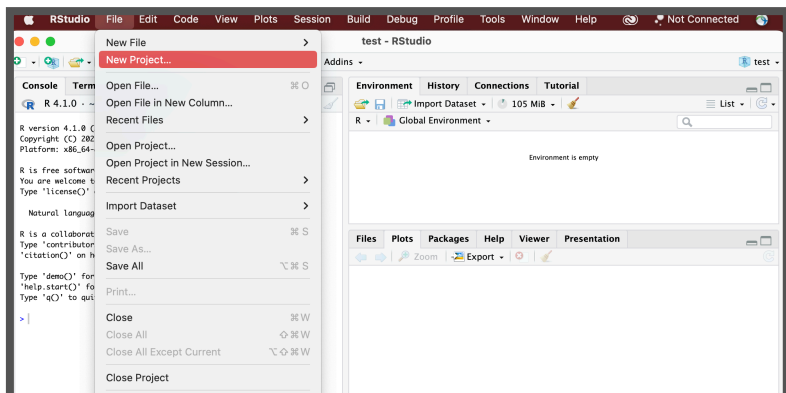
+



Git & RStudio

1. Local R project with Git

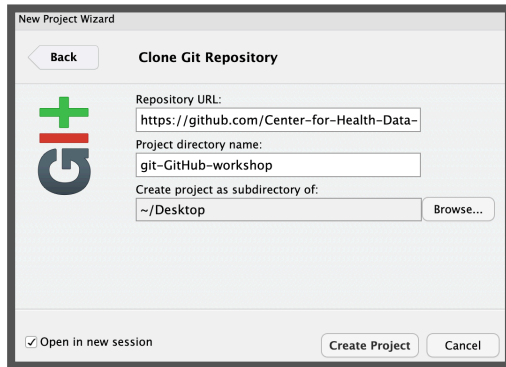
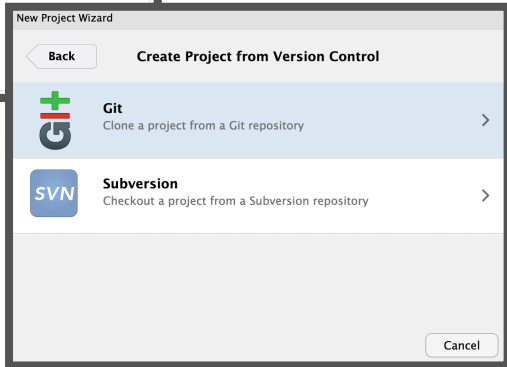
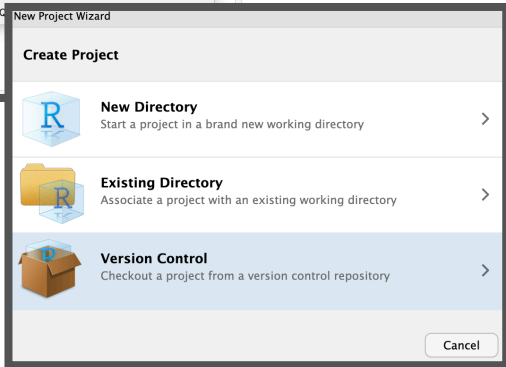
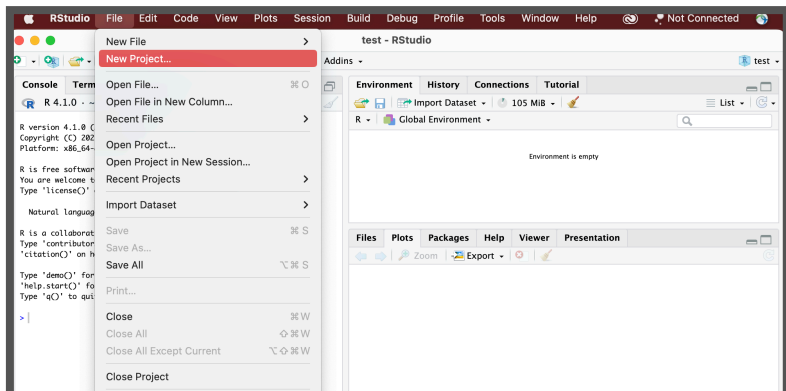
- `usethis::use_git()` to initiate Git
- `usethis::use_github()` to connect current RStudio project to GitHub (this will create new repository)



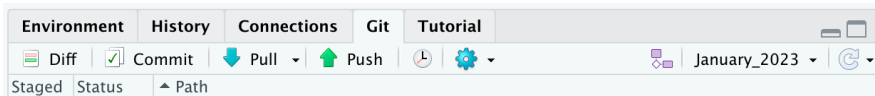
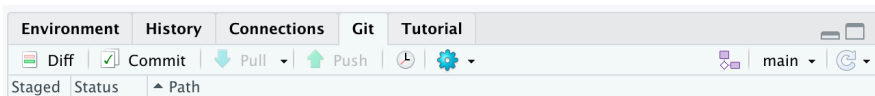
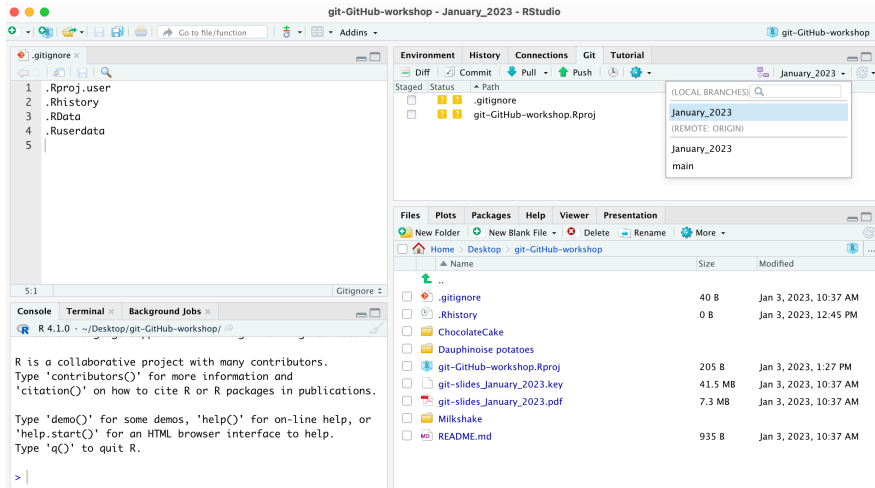
Git & RStudio

2. New R project from the GitHub repository

- Copy the GitHub repository HTTPS url
- In Rstudio, File > New Project..., select Version Control, choose Git
- Provide the repository HTTPS link you just copied



Git & RStudio

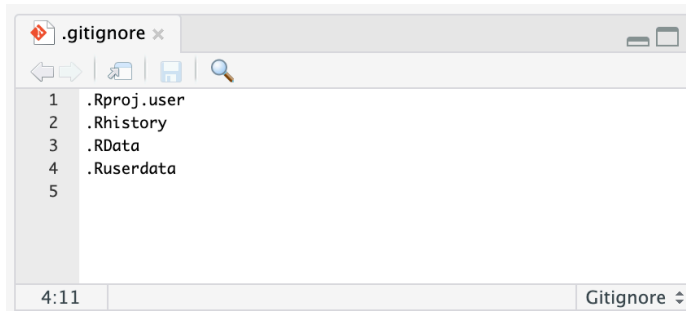


RStudio's git panel contains a subset of commands from git.

You can use more complex commands from Shell or terminal

- Make local changes, stage and commit
- Create and switch between branches
- Changes to scripts can be reverted, and pushed to remote (e.g. GitHub)
- Click the green “Push” button to send your local changes to GitHub.

Ignore files from version control



When starting a new project in RStudio, it will always add a file `.gitignore` if it does not already exist and add some initial files to ignore.

File or folder names in the `.gitignore` will not be taken into account by git or sync to GitHub

Other examples of files you would potentially want to ignore:

- Sensitive information (passwords,...)!
- Binary files. Git works very well with text files, but not with binary files such as `.Rdata`.
- Files > 50MB. Git is specifically made for code (e.g. `.R`) and does not intend to track all changes in large data files (though there are options for this).
- A temp/ folder inside your project with 'disposable' content, e.g. draft, test or temp.

Configure Git

- You want git to know who you are so it can associate your changes with you. If you haven't configured Git already you can do it from within RStudio.
- Via the usethis package in R:

```
> install.packages("usethis")  
> library(usethis)  
> usethis::use_git_config(user.name="Jane Doe", user.email="jane@example.org")
```

- `edit_git_config()` function will open your gitconfig file. Add your name and email and close this.

```
> edit_git_config()
```

Personal Access Token & RStudio

- How to Connect RStudio Projects with GitHub Repository?

Your Personal Access Token will need to be stored in a way RStudio can access it to connect to your GitHub account.

- There are a couple of ways to get your PAT into the Git credential store:
 - Call an R function to explicitly store (or update) your credentials.
 - Do something in RStudio that triggers a credential challenge.

Personal Access Token & RStudio

- **"gitcreds" package**

If you've installed "usethis", you will already have gitcreds installed

- Call `gitcreds::gitcreds_set()`.

If you don't have a PAT stored already, it will prompt you to enter your PAT. Paste!

- If you already have a stored credential, it will reveal this and will let you inspect it.

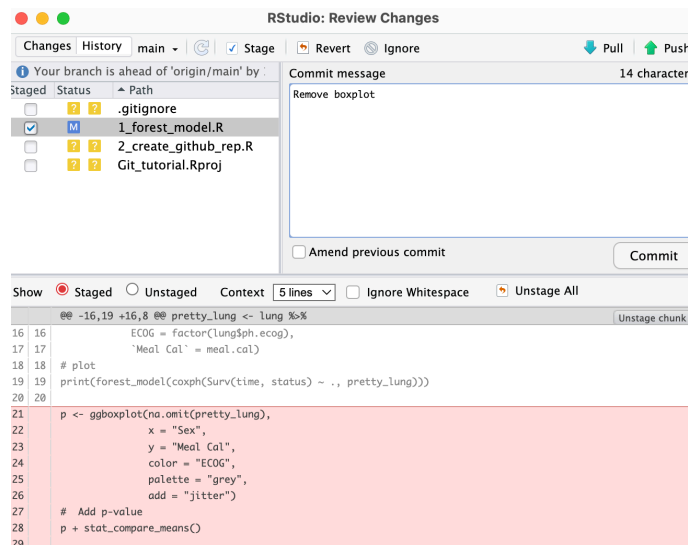
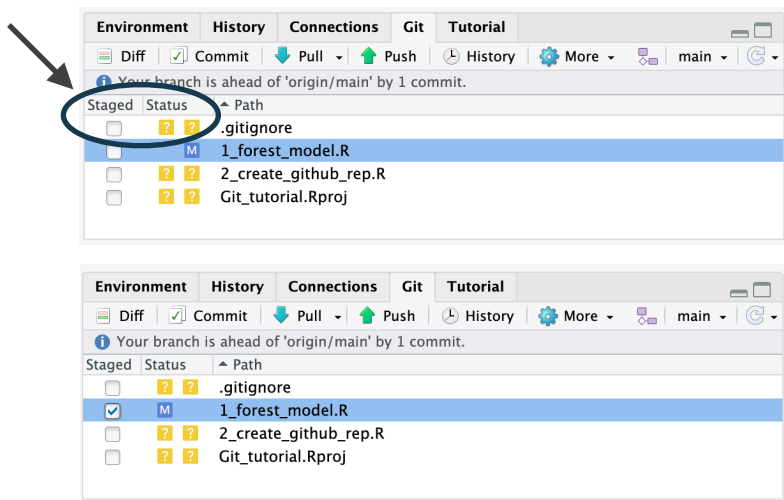
- You'll enter your GitHub username and the Personal Access Token as your password (NOT your GitHub password).

```
>install.packages("gitcreds")
>library(gitcreds)
>gitcreds_get()
#You can check that you've stored a
#credential with gitcreds_get():
```

```
>gitcreds::gitcreds_set()

? Enter password or token:
ghp_XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
XXXXXX
-> Adding new credentials...
-> Removing credentials from
cache...
-> Done.
```

Stage and Commit



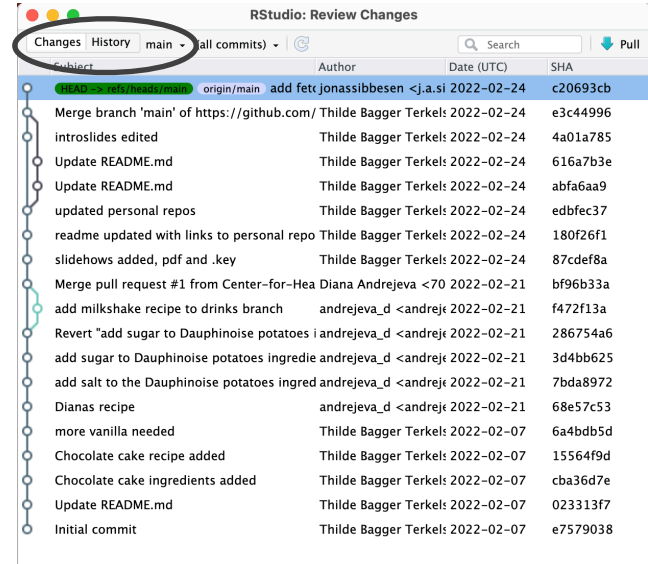
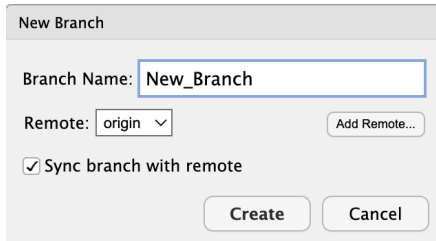
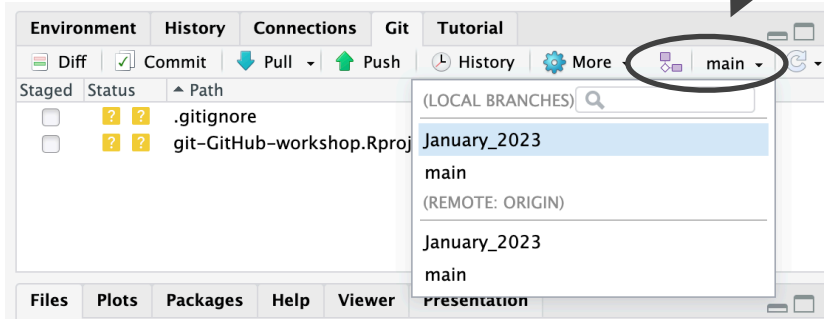
- Added
- Deleted
- Modified
- Renamed
- Untracked

The status of the new files/changes

By checking the box and clicking commit, we add the file and are able to commit this with a commit message.

THEN finally we push to GitHub!

Branching



RStudio: Review Changes

Changes History main (all commits) Search Pull

Subject	Author	Date (UTC)	SHA
HEAD -> refs/heads/main origin/main add fetr	jonassibbesen <j.a.si	2022-02-24	c20693cb
Merge branch 'main' of https://github.com/	Thilde Bagger Terkels	2022-02-24	e3c44996
introslices edited	Thilde Bagger Terkels	2022-02-24	4a01a785
Update README.md	Thilde Bagger Terkels	2022-02-24	616a7b3e
Update README.md	Thilde Bagger Terkels	2022-02-24	abfa6aa9
updated personal repos	Thilde Bagger Terkels	2022-02-24	edbfec37
readme updated with links to personal repo	Thilde Bagger Terkels	2022-02-24	180f26f1
slidehows added, pdf and .key	Thilde Bagger Terkels	2022-02-24	87cdef8a
Merge pull request #1 from Center-for-Hea	Diana Andrejeva <70	2022-02-21	bf96b33a
add milkshake recipe to drinks branch	andrejeva_d <andrej	2022-02-21	f472f13a
Revert "add sugar to Dauphinoise potatoes i	andrejeva_d <andrej	2022-02-21	286754a6
add sugar to Dauphinoise potatoes ingredie	andrejeva_d <andrej	2022-02-21	3d4bb625
add salt to the Dauphinoise potatoes ingred	andrejeva_d <andrej	2022-02-21	7bda8972
Dianas recipe	andrejeva_d <andrej	2022-02-21	68e57c53
more vanilla needed	Thilde Bagger Terkels	2022-02-07	6a4bdb5d

Commits 1-19 of 19

README.md

View file @ abfa6aa9

```
@@ -1,4 +1,4 @@
1 # git-GitHub-workshop-2022
1 ## git-GitHub-workshop-2022
2 2
3 3 This repository is used for the HeadS git and GitHub workshop 2022.
4 4 Workshop participants should clone the repo after which they can use it to complete workshop
   exercises associated. The shared repo is made for hosting cooking recipies which are made
   locally by participants and then pushed to remote for everyone to view.
@@ -14,7 +14,7 @@ josi@di.ku.dk
14 14 Diana Andrejev, Data Scientist
15 15 andrejeva@sund.ku.dk
16 16
17 17 For the workshops last exercise we will be working with three private repositorys to learn to
   work with requests:
17 # For the workshops last exercise we will be working with three private repositorys to learn to
   work with requests:
18 18
```

History

Environment History Connections Git Tutorial

Diff Commit Pull Push January_2023

Staged Status Path

- Notice the presence of History twice:
- Git history vs R command history
- You can check the history online, using the terminal as well as in Rstudio.
- Click on history in the **Git panel**

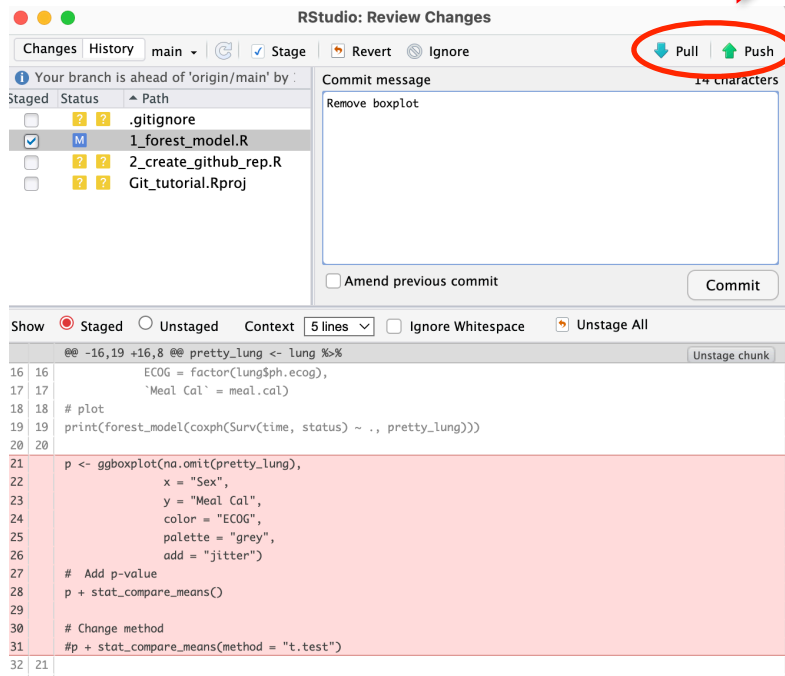
Interact with GitHub

Pull updates from GitHub

- Using the down arrow button, RStudio goes to the GitHub repo, grabs the most recent code and brings it into your local editor.
- Pulling regularly is extremely important if you're collaborating.

Push changes to GitHub

- Rstudio gives you a warning about the status of your local commits versus those stored on GitHub.
- After committing, we now have a push button (the up arrow) on RStudio.

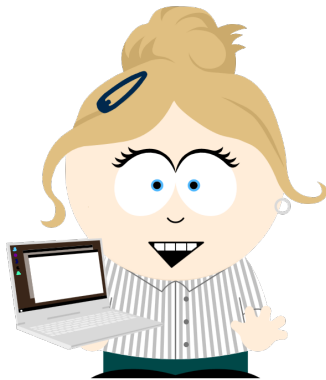


Git & RStudio

Example

1

Let's go to RStudio



Thank you for today
See you at HeaDS

IN CASE OF FIRE 

 git commit

 git push

 git -tf out